



Tech Talk #47

Beyond the Search Bar Engineering the Future of *Conversational E-commerce*



Riccardo Belluzzo
Senior Research Engineer | Language Intelligence Team

allegro

Rozwiń talent

z e-Xperience!



E-XPERIENCE



6 miesięcy
praktycznych zadań
(lipiec-grudzień)



Umowa o pracę



Wsparcie buddy'ego

**Rekrutacja
wystartowała! Aplikuj
na jobs.allegro.eu**

Bądź na bieżąco, zaobserwuj nas na social media



INSTAGRAM



FACEBOOK

@allegrodobrzetubyc @Allegro Tech
@Allegro Jobs

Kogo poszukujemy w programie e-Xperience?



→ **Junior Automation Engineer**

→ **Junior Data Engineer/Data Scientist**

→ **Junior Software Engineer (Java/Kotlin, Python, Mobile)**

Pełna lista stanowisk wraz ze
szczegółami na Allegro Jobs

Polecamy zeskanować kod QR 🔥



AI HUB

- **70+ experts**
(Engineering, Data,
Research & Product)
- **Impactful AI
applications &
cutting edge
research**
(ml.allegro.tech)

Product Analyst

Poznań, Nowy Rynek

Data Scientist – Central Query Understanding

Poznań, Nowy Rynek

Senior Research Engineer – Learning to Rank

Warszawa, Fabryka Norblina

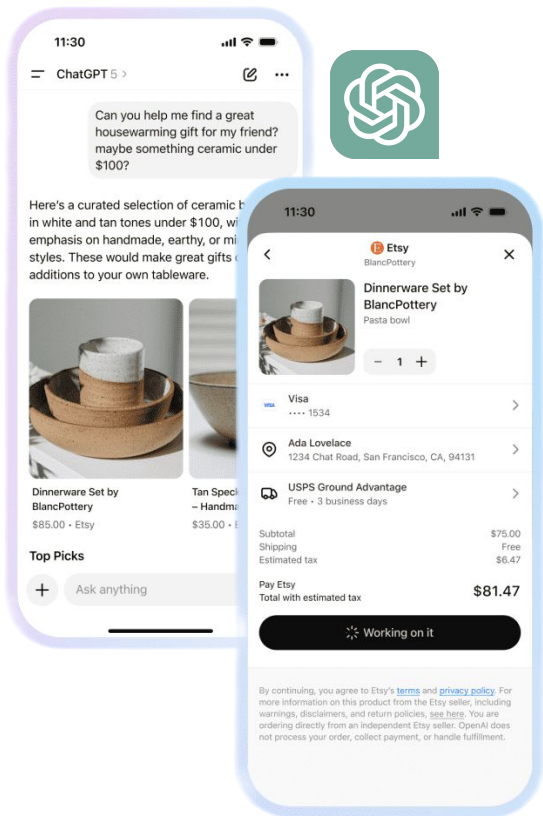
Senior Software Engineer (Java / Kotlin)

Poznań, Nowy Rynek

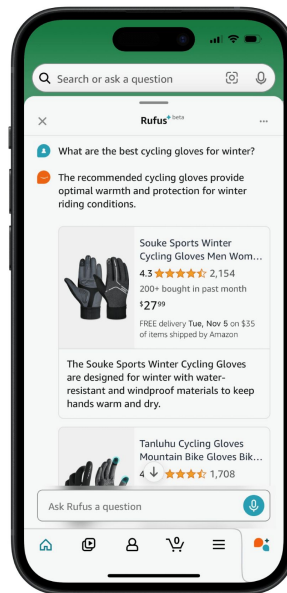


allegro

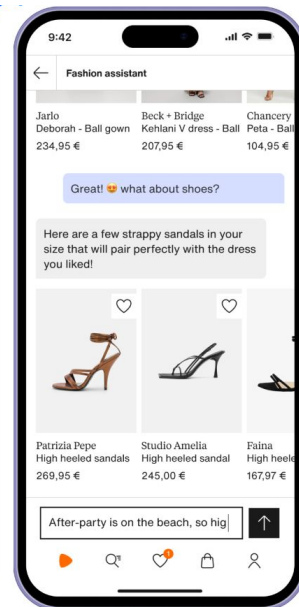
Chatbots everywhere



Ask Rufus



zalando



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Conversational e-commerce is coming

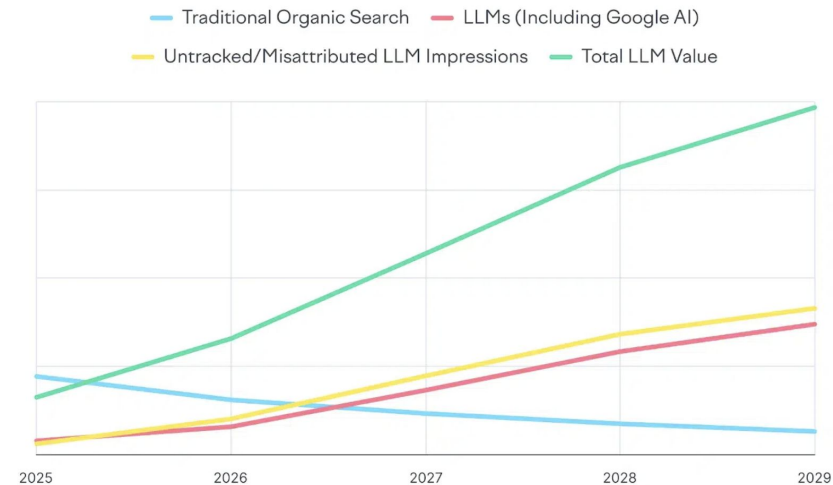
ChatGPT Shopping is here – and it's changing ecommerce SEO rules

Source: October 2025 - [SearchEngine Land](#)

ChatGPT referrals to retailers' apps increased 28% year-over-year, says report

Source: December 2025 - [Tech Crunch](#)

Estimated LLM vs. Organic Search Value



semrush.com

SEMRUSH

Introduction

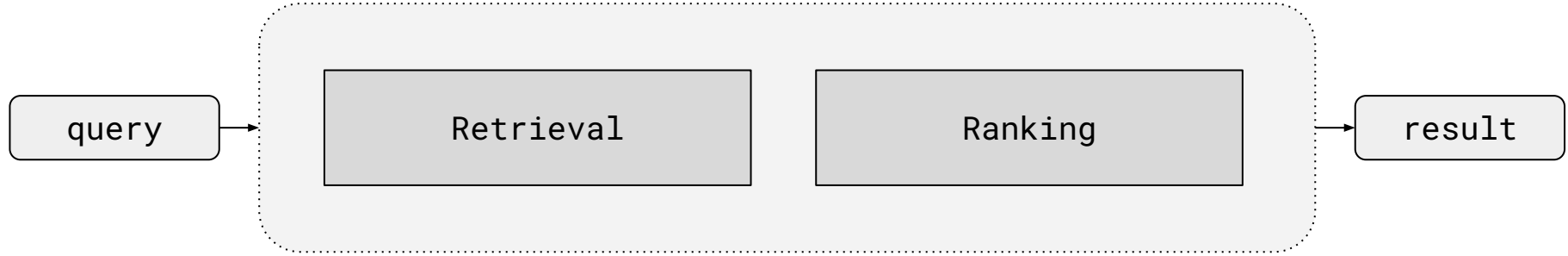
Conversational E-commerce
through the lens of *search*



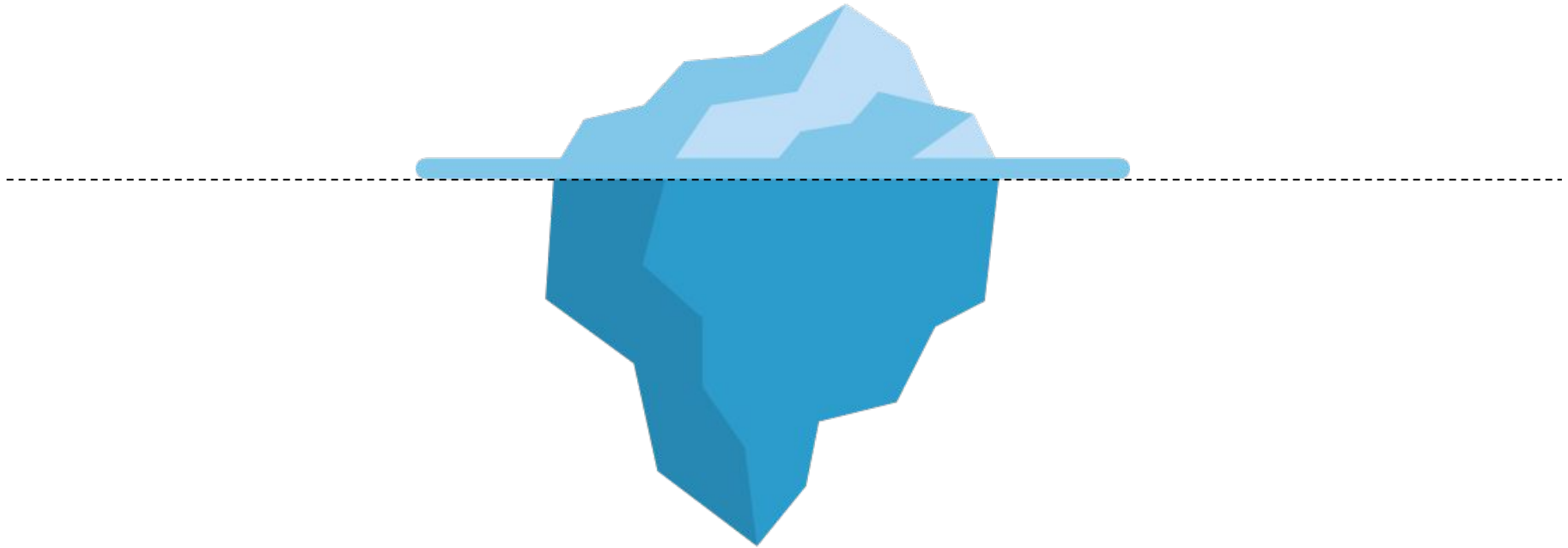
The backbone of modern e-commerce systems



The backbone of modern e-commerce systems



The Semantic Gap

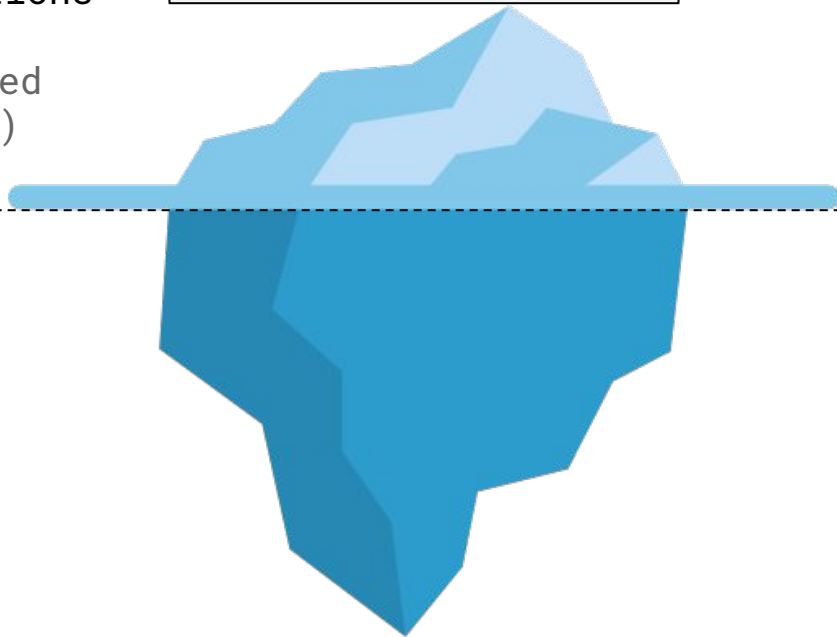


The Semantic Gap

- Product titles
- Product parameters
- Product descriptions

Explicit Information

(signals normally used
to train algorithms)



The Semantic Gap

- Product titles
- Product parameters
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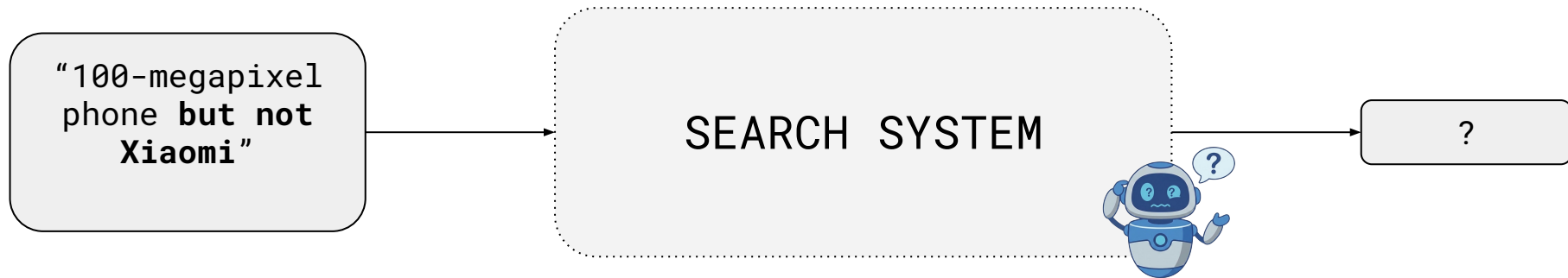
Explicit Information

Implicit Information

- What is this product capable of?
- Who is this product suitable for?
- What people think about it?

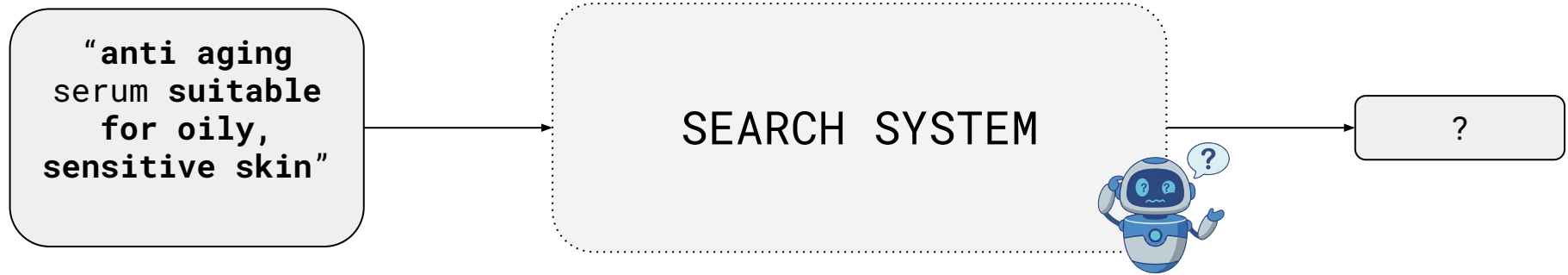
(signals humans care about!)

Semantic Gap → Search pitfalls



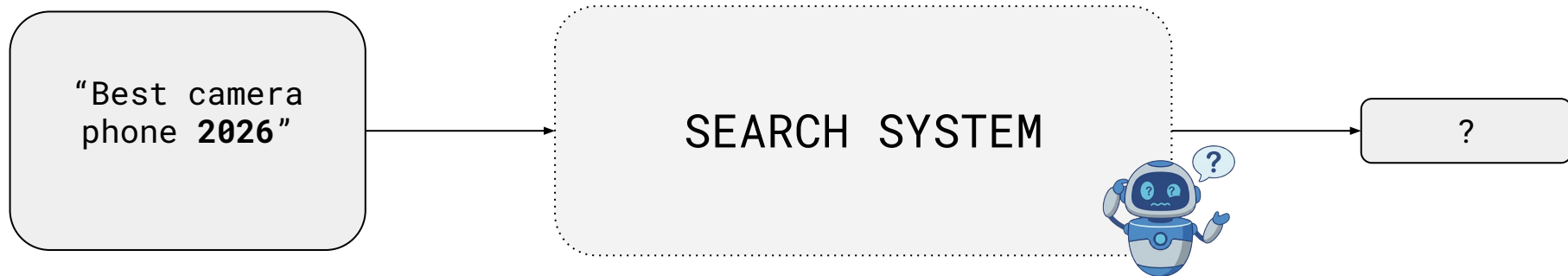
Failure in handling **strong semantic constraints**.

Semantic Gap → Search pitfalls



Failure in handling **multi-attribute implicit needs**.

Semantic Gap → Search pitfalls



Static indexing - impossibility of reasoning on up-to-date knowledge.

How to mitigate the Semantic Gap?

How to mitigate the Semantic Gap?

- All approaches aimed to improve Retrieval and Ranking models are valid. For example:
 - Better training data
(more aligned with implicit needs);
 - Multimodality;
 - etc.

How to mitigate the Semantic Gap?

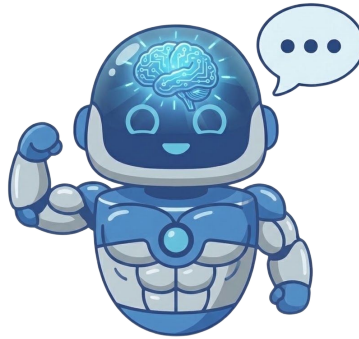
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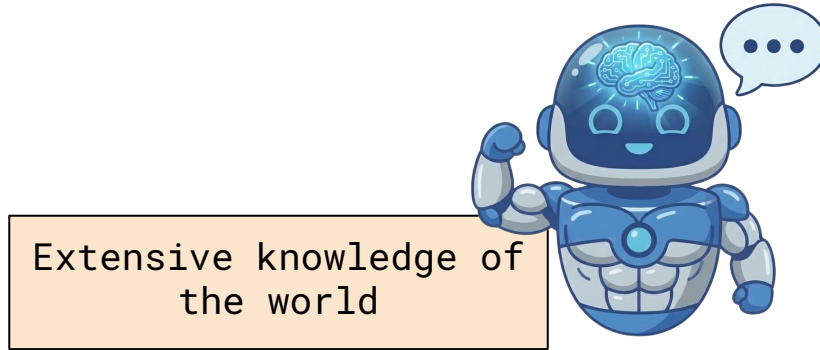
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 - etc.
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Is there an easier way to try to mitigate the problem?

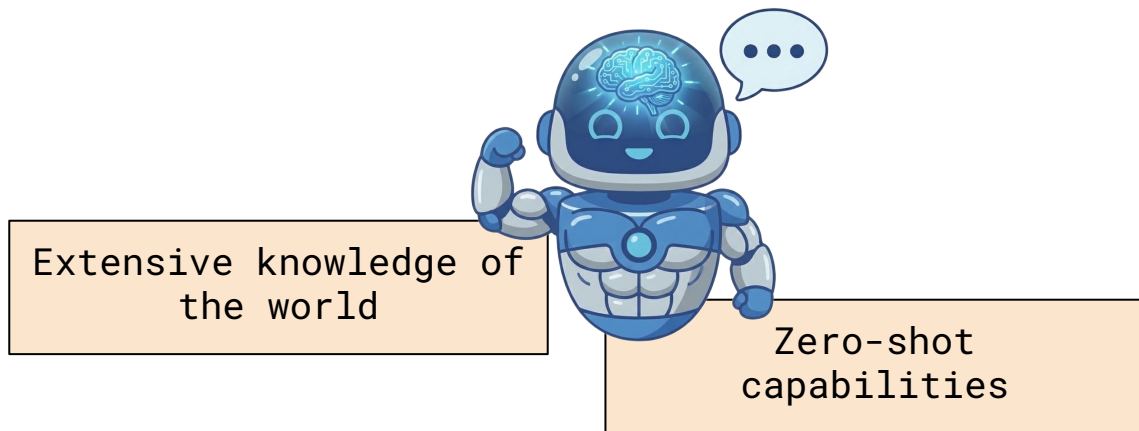
LLMs enter the game



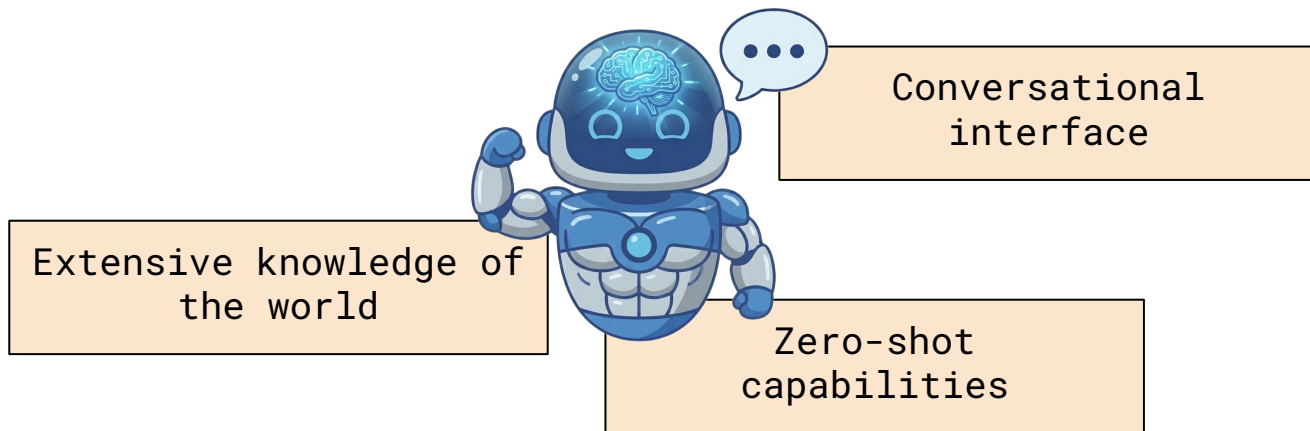
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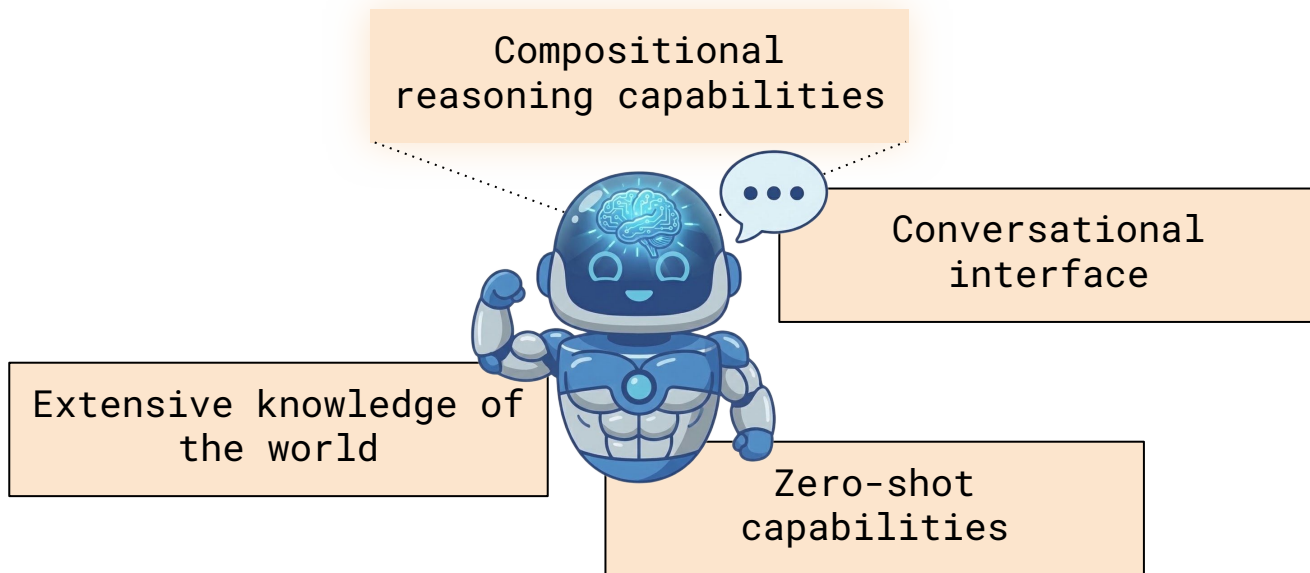


LLMs enter the game



LLMs enter the game

"The ability of reasoning on, decomposing, and refine search queries through multi-turn conversations."



Building the Conversational E-commerce System Specifications



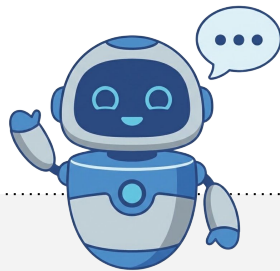
Conversational Search System Specifications



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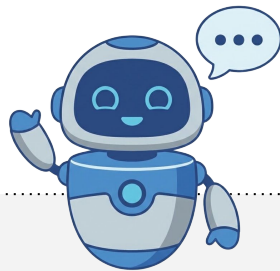
Conversational Search System Specifications



"100-mega
pixel
phone but
not
Xiaomi"

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1. Guidance Module



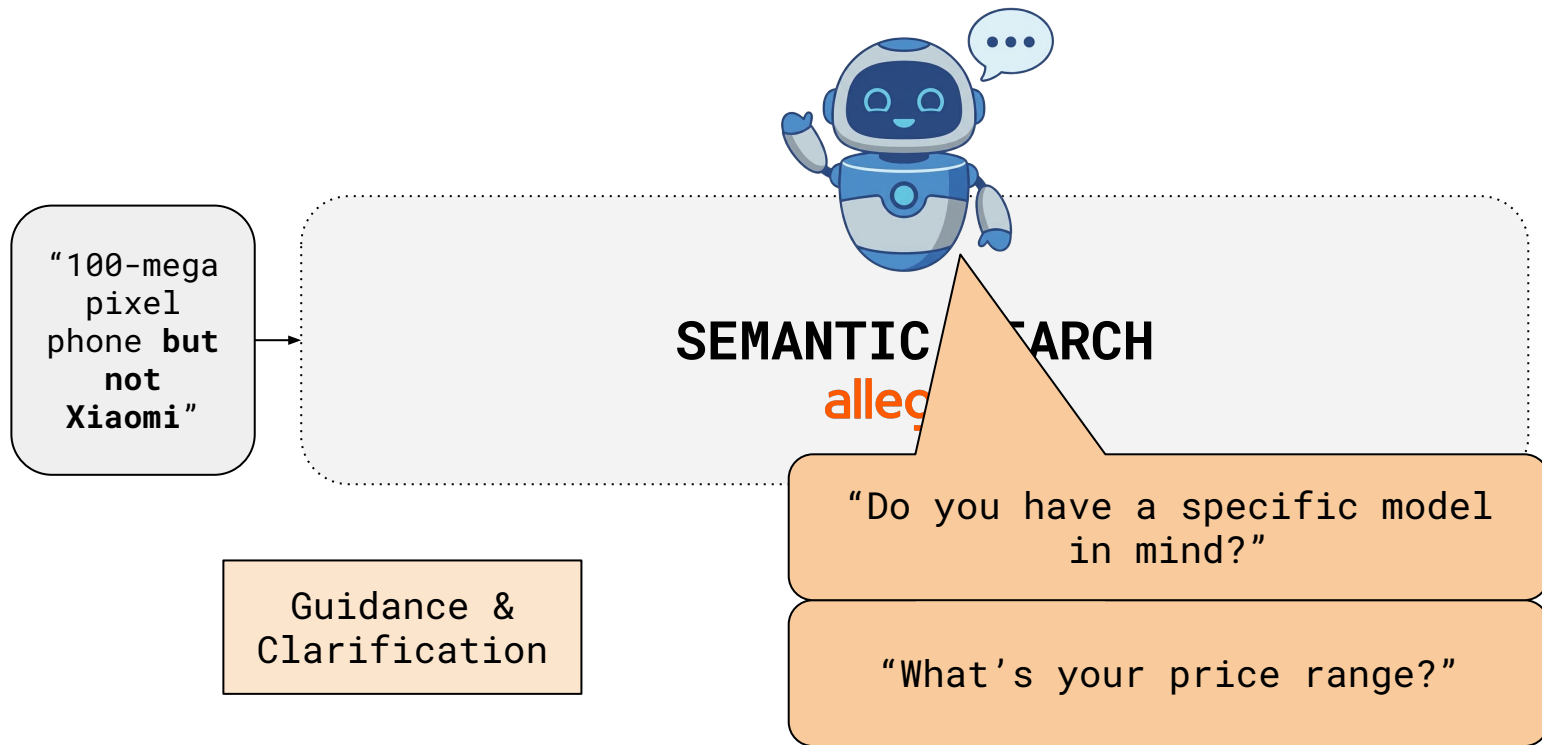
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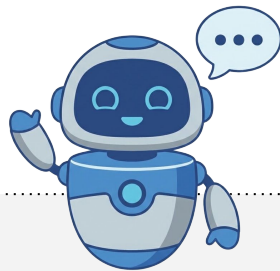
SEMANTIC SEARCH
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Guidance &
Clarification

1. Guidance Module



2. Query Understanding Module



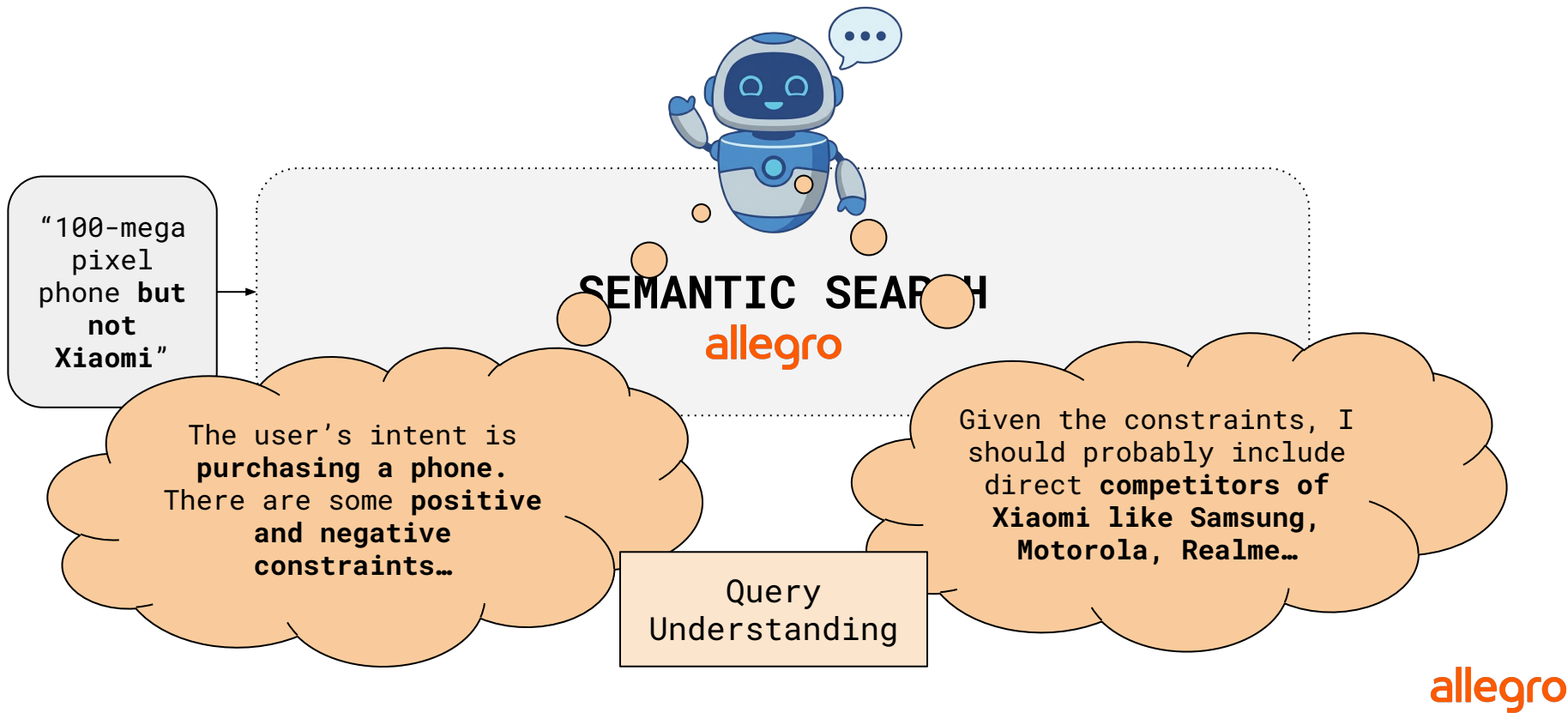
"100-mega
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SEMANTIC SEARCH
allegro

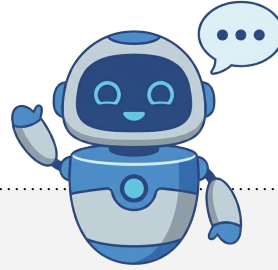
Guidance &
Clarification

Query
Understanding

2. Query Understanding Module



3. Explainability Module



"100-mega
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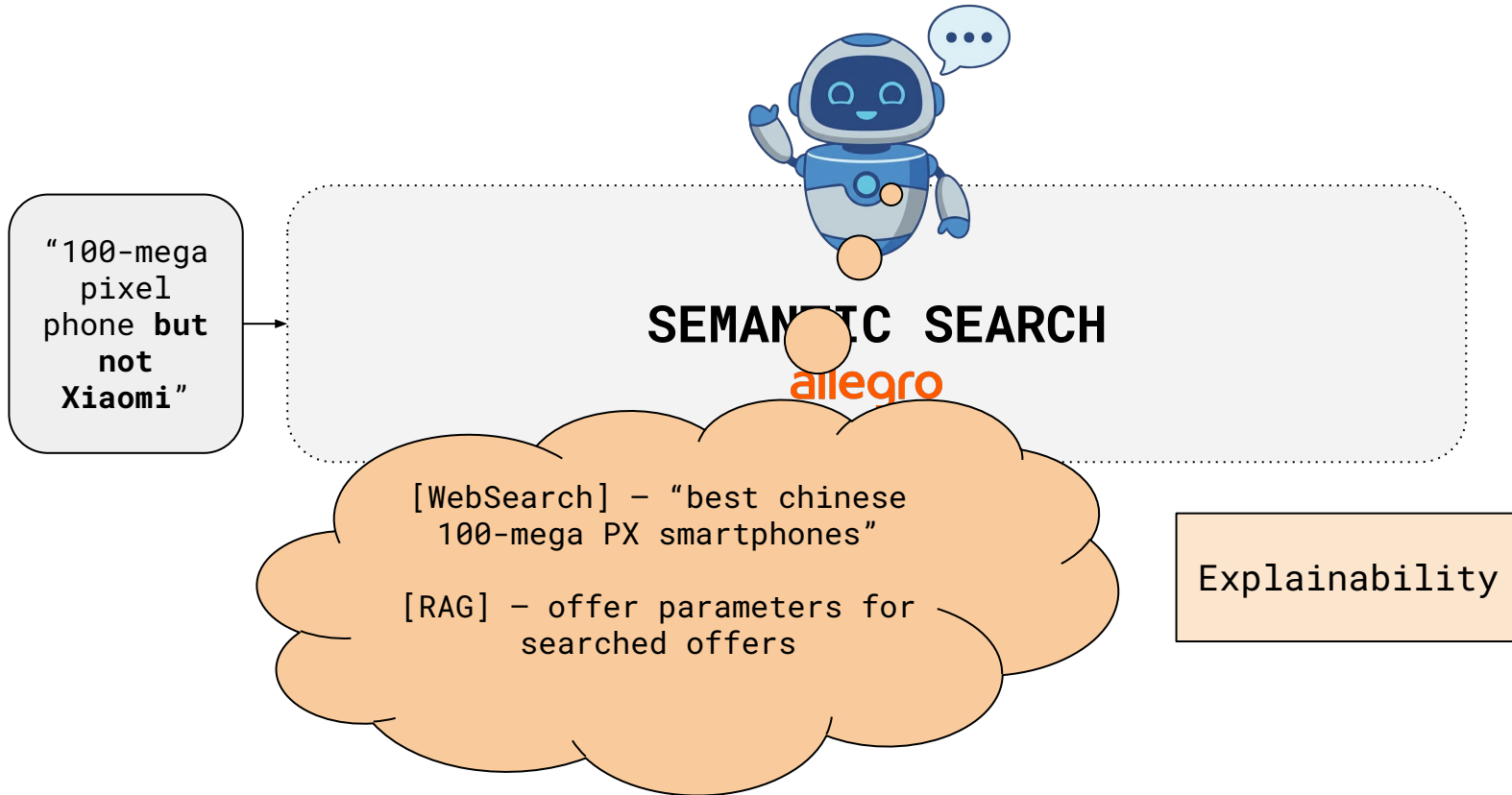
SEMANTIC SEARCH
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Guidance &
Clarification

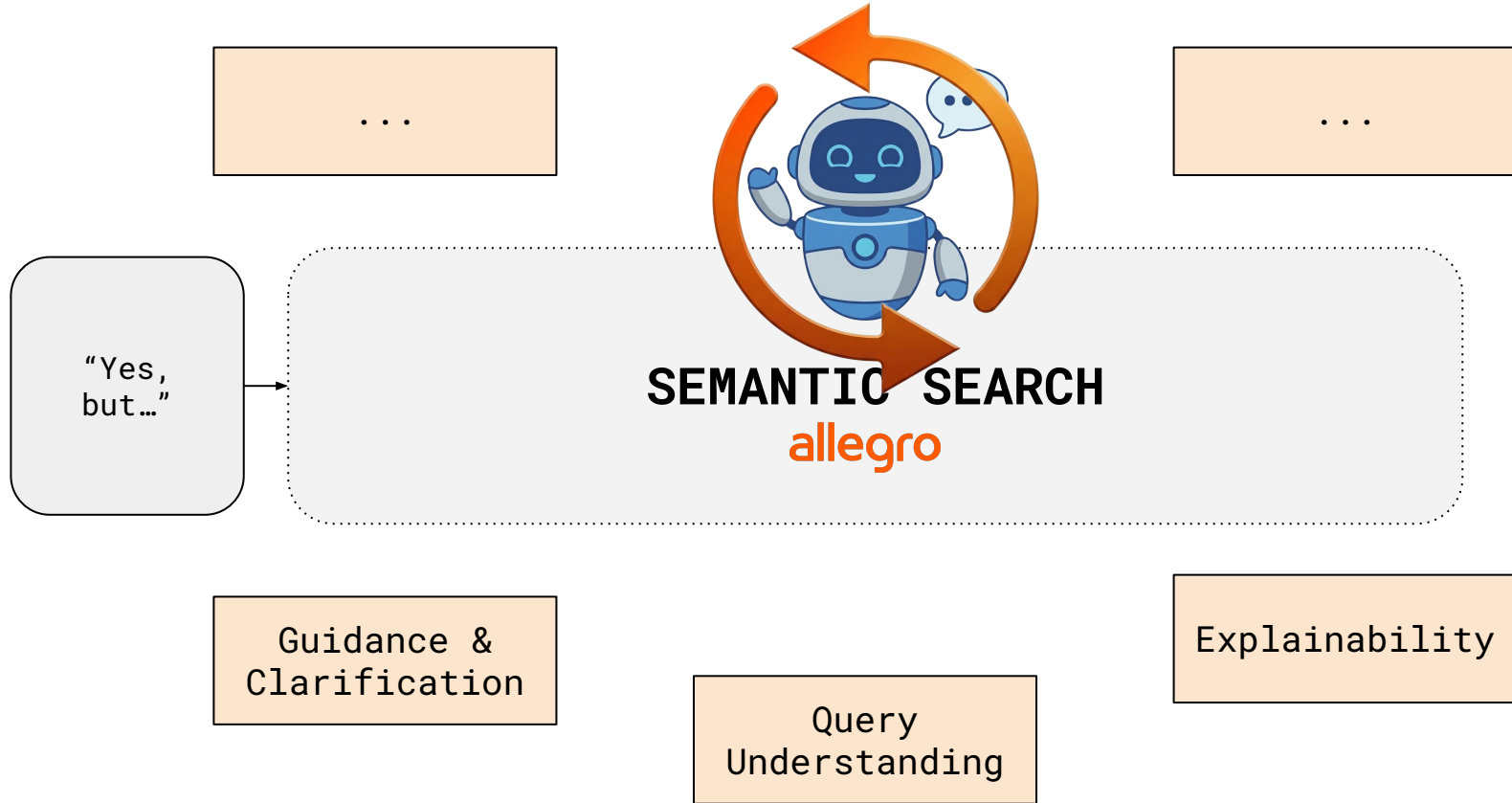
Query
Understanding

Explainability

3. Explainability Module



4. Multi-turn Refinement



System specifications

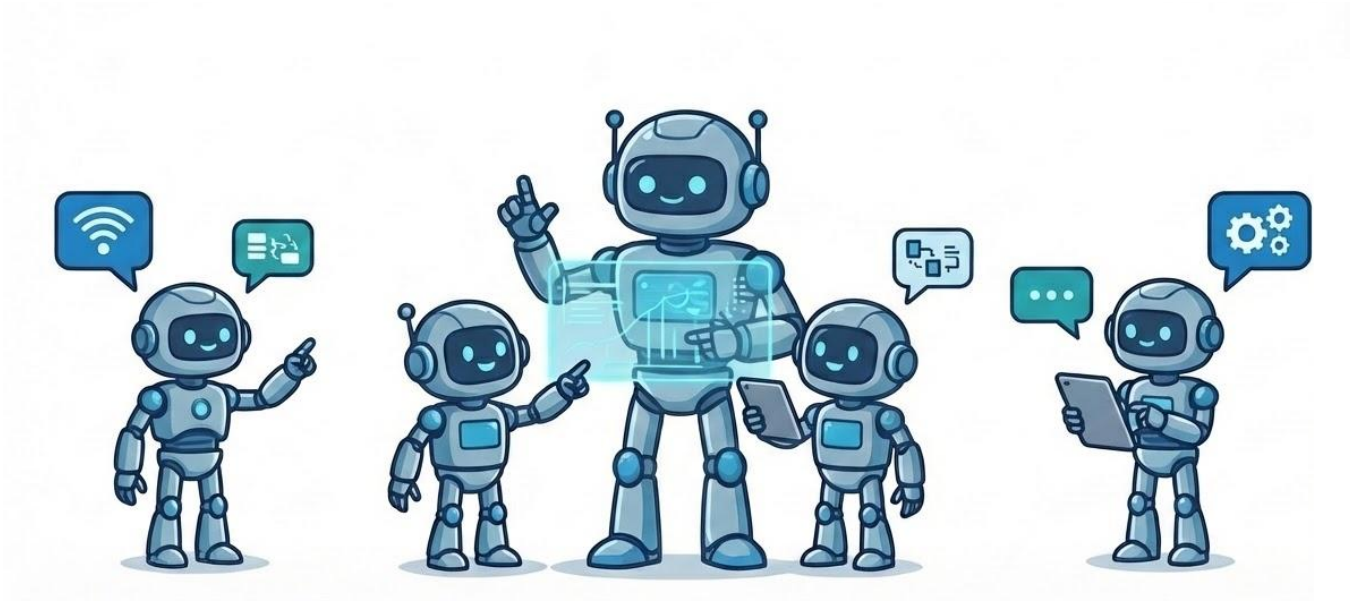
1. **Decomposition of the problem into specialized components;**
2. **Multi-turn refinement of the results:**
 - Sequential and parallel operations;
 - Ability to “self-reflect”;
 - Shared “memory” of what is happening.

System specifications

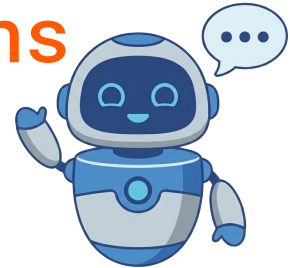
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Does it sound familiar?

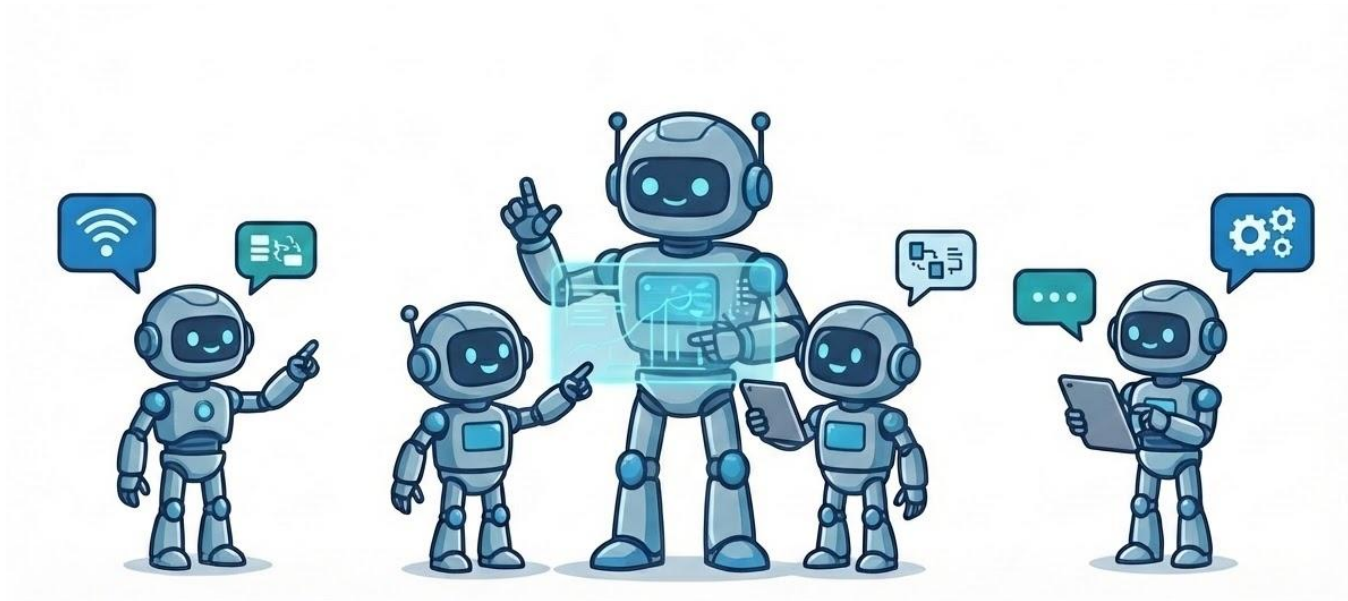
Conversational E-commerce as a Multi-agent Collaborative Environment



Building the Conversational E-commerce Multi-agent Systems Foundations

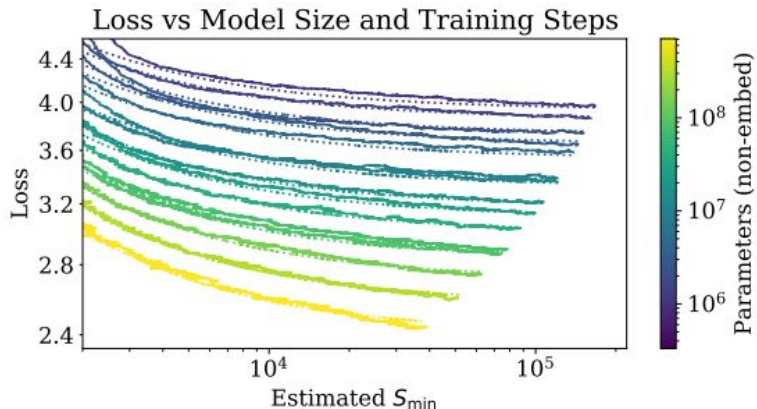
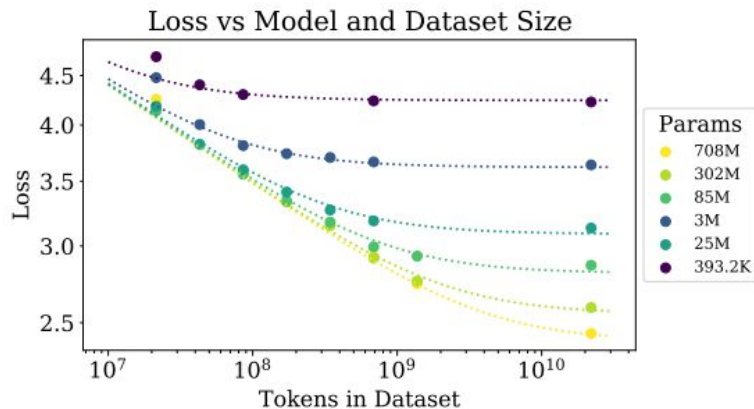


Theoretical foundations for building Multi-agent systems (?)



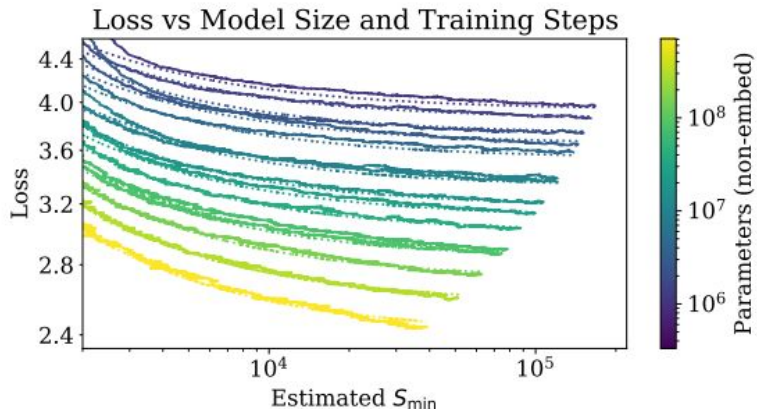
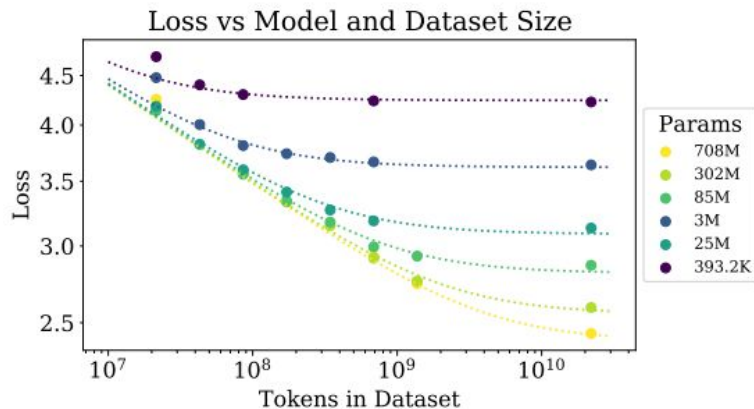
Scaling Laws

In Deep Learning, practitioners can take advantage of theoretical foundations when designing a new model.



Scaling Laws

In Deep Learning, practitioners can take advantage of theoretical foundations when designing a new model.



Do we have something similar for building Multi-agent systems?

Scaling Laws for Agents

Google Research Google DeepMind

Towards a Science of Scaling Agent Systems

Yubin Kim^{1,3,†}, Ken Gu¹, Chanwoo Park³, Chunjong Park², Samuel Schmidgall², A. Ali Heydari¹, Yao Yan¹, Zhihan Zhang¹, Yuchen Zhuang², Yun Liu¹, Mark Malhotra¹, Paul Pu Liang³, Hae Won Park³, Yuzhe Yang¹, Xuhai Xu¹, Yilun Du¹, Shwetak Patel¹, Tim Althoff¹, Daniel McDuff¹ and Xin Liu^{1,†}

¹Google Research, ²Google DeepMind, ³Massachusetts Institute of Technology, [†]Corresponding Author

Agents, language model (LM)-based systems that are capable of reasoning, planning, and acting are becoming the dominant paradigm for real-world AI applications. Despite this widespread adoption, the principles that determine their performance remain underexplored, leaving practitioners to rely on heuristics rather than principled design choices. We address this gap by deriving quantitative *scaling principles* for agent systems. We first formalize a definition for agentic evaluation and characterize scaling laws as the interplay between agent quantity, coordination structure, model capability, and task properties. We evaluate this across four diverse benchmarks: FINANCE-AGENT, BROWSECOMP-PLUS, PLAN-CRAFT, and WORKBENCH, spanning financial reasoning, web navigation, game planning, and workflow execution. Using five canonical agent architectures (Single-Agent System and four Multi-Agent Systems: Independent, Centralized, Decentralized, Hybrid), instantiated across three LLM families, we perform a controlled evaluation spanning 180 configurations, standardizing tools, prompt structures, and token budgets to isolate architectural effects from implementation confounds. We derive a predictive model using empirical coordination metrics, including efficiency, overhead, error amplification, and redundancy, that achieves cross-validated $R^2=0.524$, enabling prediction on unseen task domains by modeling task properties rather than overfitting to a specific dataset. We identify three dominant effects: (1) a *tool-coordination trade-off*: under fixed computational budgets, tool-heavy tasks suffer disproportionately from multi-agent overhead. (2) a *capability saturation*: we observe that coordination yields diminishing or negative returns ($\beta=-0.404$, $p<0.001$) once single-agent baselines exceed an empirical threshold of $\sim 45\%$. (3) *topology-dependent error amplification*: independent agents amplify errors 17.2 $\times$ through unchecked propagation, while centralized coordination contains this to 4.4 $\times$. Crucially, coordination benefits are task-contingent. Centralized coordination improves performance by 80.8% on parallelizable tasks like financial reasoning, while decentralized coordination excels on dynamic web navigation (+9.2% vs. +0.2%). Yet for sequential reasoning tasks, every multi-agent variant we tested degraded performance by 39–70%. The framework predicts the optimal coordination strategy for 87% of held-out configurations. Out-of-sample validation on GPT-5.2, released after our study, achieves MAE=0.071 and confirms four of five scaling principles generalize to unseen frontier models, providing a quantitatively predictive framework for *agentic scaling* based on measurable task properties.

arXiv:2512.08296v2 [cs.AI] 17 Dec 2025

Topology Patterns for building Multi-agent systems

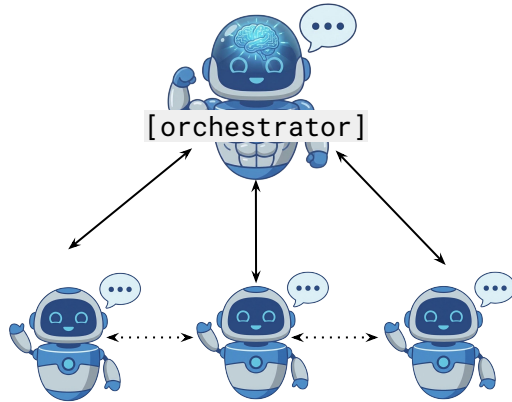
Single Agent System [SAS]



- A single agent that does everything, sequentially.
- “Think step by step”...
- **Sequential** operations. No communication overhead.

Topology Patterns for building Multi-agent systems

CENTRALIZED

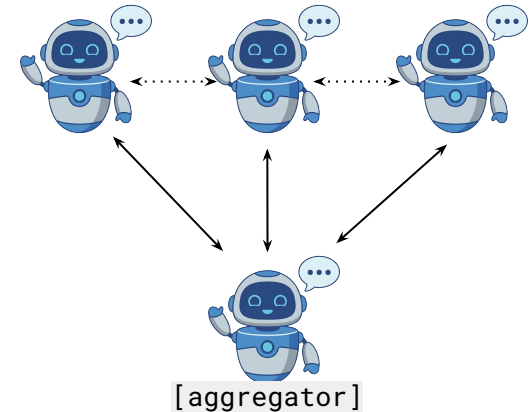


Multi Agent System [MAS]

- **Information fusion** with (CENTRALIZED) or without (DECENTRALIZED) **hierarchy**;
- **Parallelism** enabled;
- Optional **peer debating**;
- More communication overhead.

◄.....► optional *peer debating*

DECENTRALIZED



Topology - Which one to choose? [*Math Edition*]

$$\begin{aligned} P = & \beta_0 + \beta_1(I - \bar{I}) + \beta_2(I - \bar{I})^2 + \beta_3 \log(1 + T) + \beta_4 \log(1 + n_a) \\ & + \beta_5 \log(1 + O\%) + \beta_6 c + \beta_7 R + \beta_8 E_c + \beta_9 \log(1 + A_e) \\ & + \beta_{10} P_{SA} + \beta_{11}(I \times E_c) + \beta_{12}(A_e \times P_{SA}) \\ & + \beta_{13}(O\% \times T) + \beta_{14}(R \times n_a) + \beta_{15}(c \times I) \\ & + \beta_{16}(E_c \times T) + \beta_{17}(P_{SA} \times \log(1 + n_a)) \\ & + \beta_{18}(I \times \log(1 + T)) + \beta_{19}(A_e \times T) + \varepsilon, \end{aligned}$$

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$$P_{SA}^* = -\frac{\hat{\beta}_4}{\hat{\beta}_{17}} \approx \frac{0.052}{0.404} = 0.129 \quad (\text{in standardized units}),$$

corresponding to raw performance ≈ 0.45 after denormalization.

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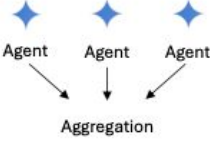
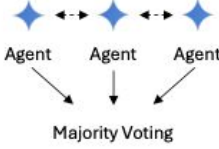
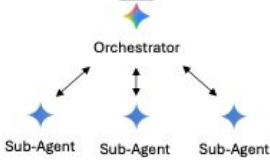
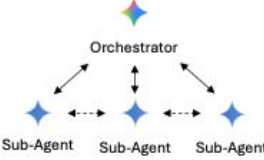
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Go Multi-agent $\leftarrow < 0.45$

$> 0.45 \rightarrow$ Single agent is enough

Topology - Which one to choose?

Characteristic	SAS	MAS (Independent)	MAS (Decentralized)	MAS (Centralized)	MAS (Hybrid)
Interaction Type	 Agent	 Aggregation	 Majority Voting	 Sub-Agent Sub-Agent Sub-Agent	 Sub-Agent Sub-Agent Sub-Agent
LLM Calls	$O(k)$	$O(nk) + O(1)$	$O(dnk) + O(1)$	$O(rnk) + O(r)$	$O(rnk) + O(r) + O(p)$
Sequential Depth	k	k	d	r	r
Comm. Overhead	0	1	$d \cdot n$	$r \cdot n$	$r \cdot n + p \cdot m$
Parallelization Factor	1	n	n	n	n
Memory Complexity	$O(k)$	$O(n \cdot k)$	$O(d \cdot n \cdot k)$	$O(r \cdot n \cdot k)$	$O(r \cdot n \cdot k + p \cdot n)$
Coordination	Sequential	Parallel + Synthesis	Sequential Debate	Hierarchical	Hierarchical + Peer
Consensus	-	Synthesis	Debate	Orchestrator	Orchestrator

* k = max iterations per agent, n = number of agents, r = orchestrator rounds, d = debate rounds, p = peer communication rounds, m = average peer requests per round. Communication overhead counts inter-agent message exchanges. Independent offers maximal parallelization with minimal coordination. Decentralized uses sequential debate rounds. Hybrid combines orchestrator control with directed peer communication.

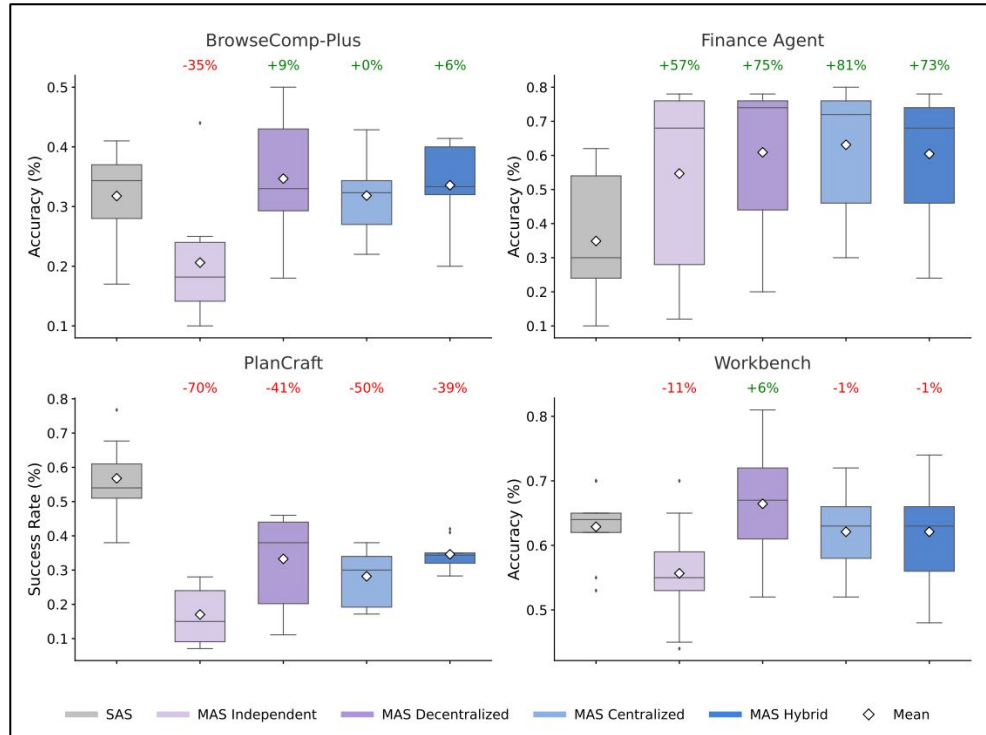
Topology - Which one to choose?

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It depends on what is the *environment* your system is gonna work.

Topology - Which one to choose?

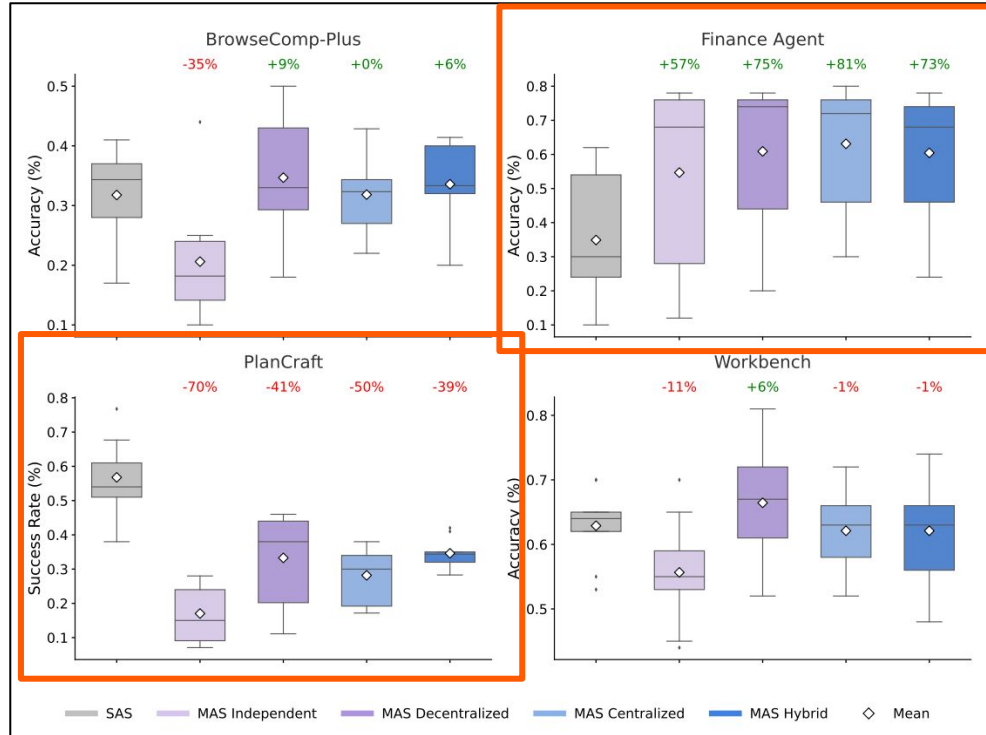
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[Towards a Science of Scaling Agent Systems](#)

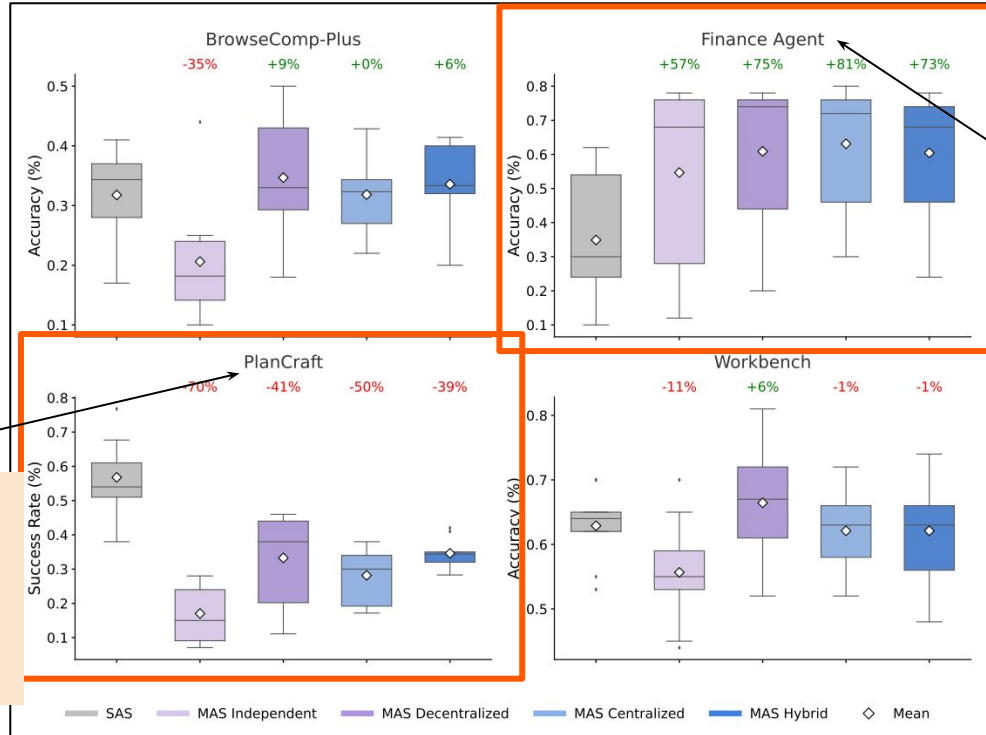
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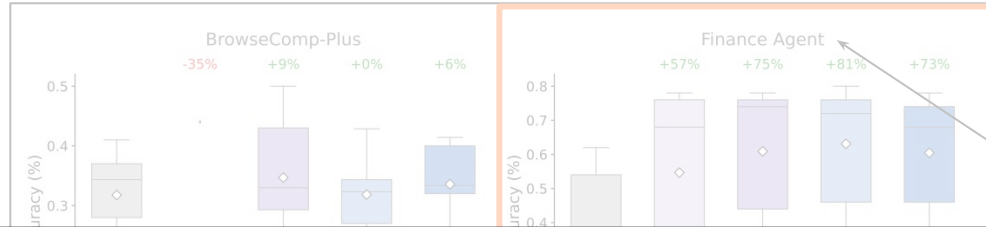


- Requires orchestration;
- High role specialization

- Very Sequential;
- No role diversity

Topology - Which one to choose?

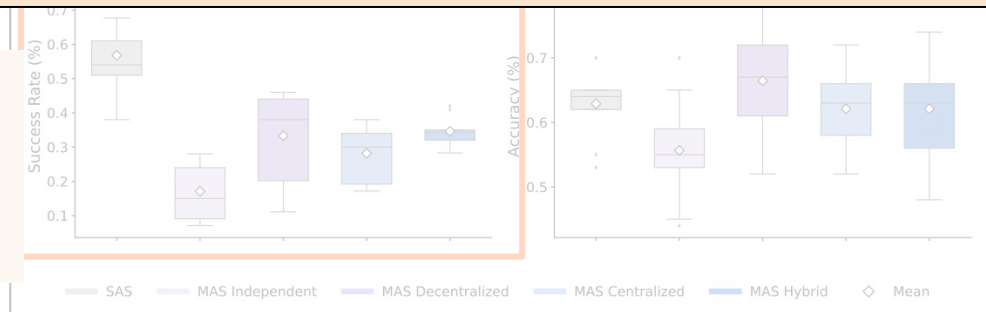
It depends on what is the *environment* your system is gonna work.



- Requires orchestration;
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So, which kind of environment is an **e-commerce** platform?

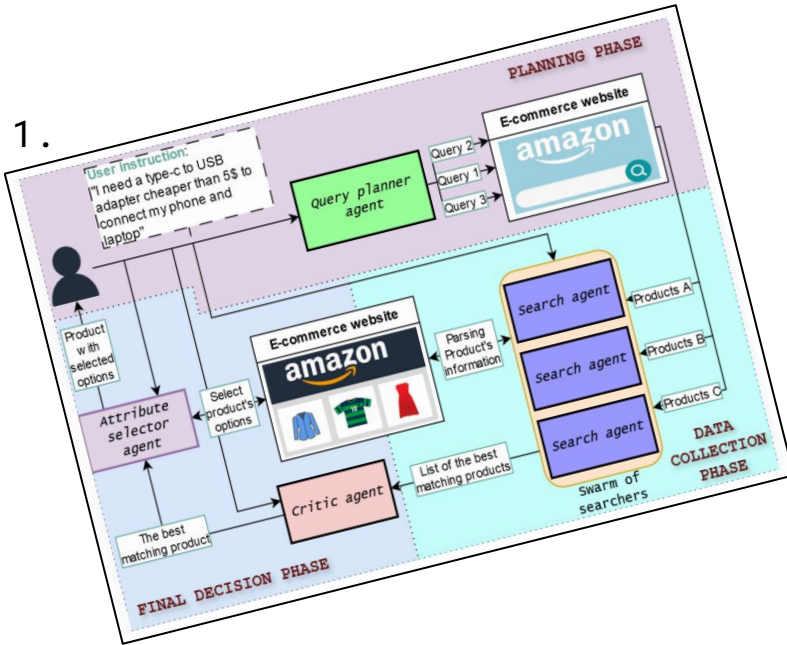
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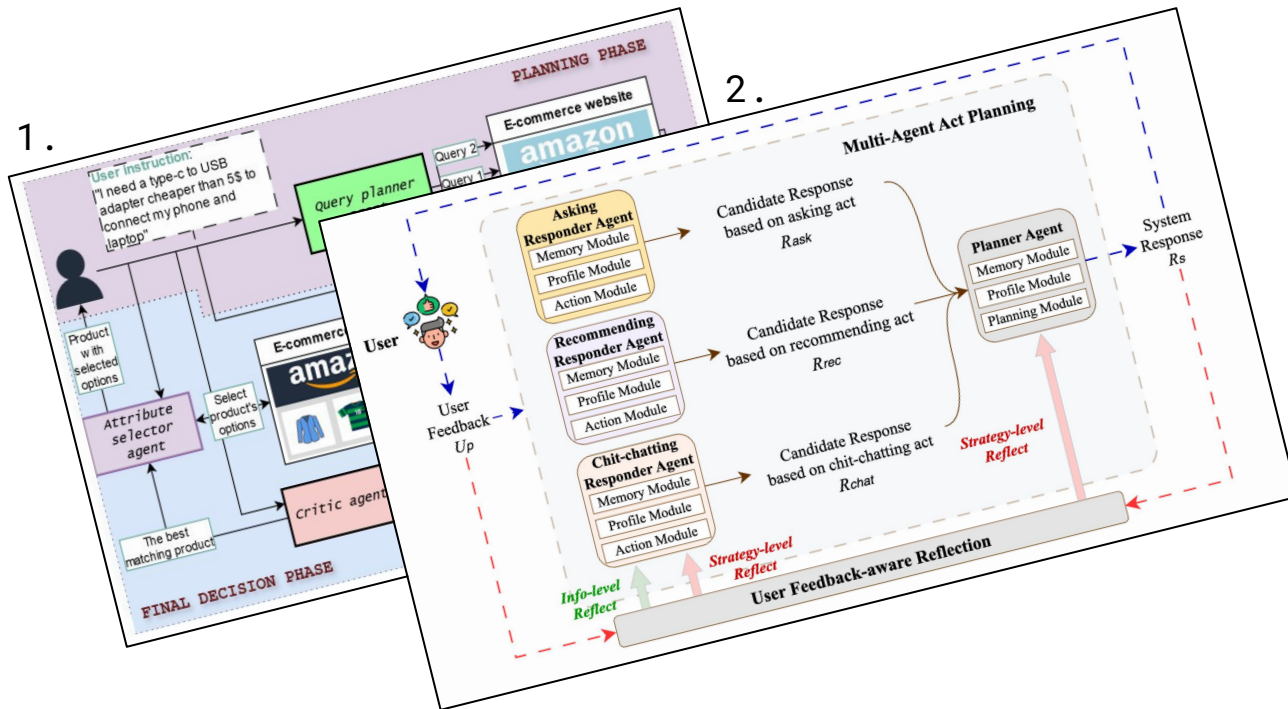
What the literature says...

What the literature says...



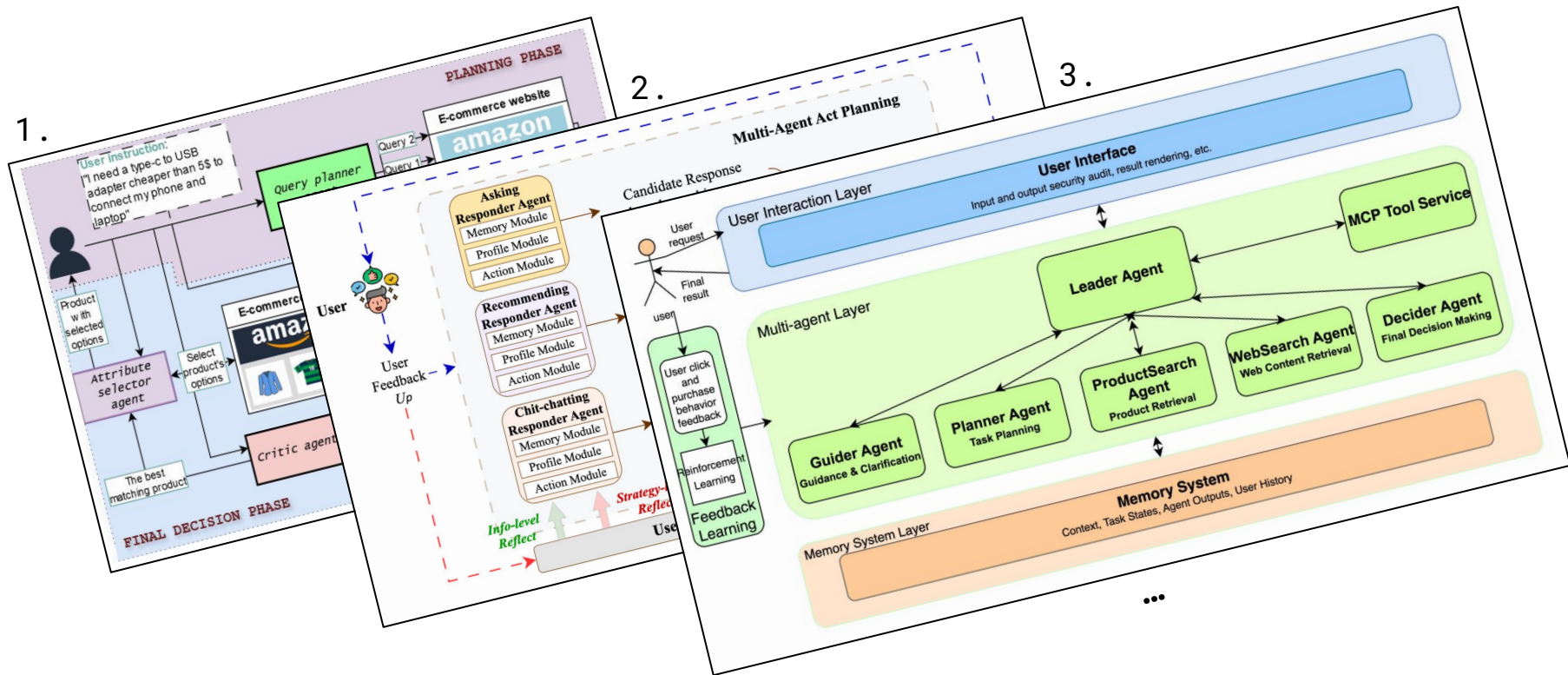
1. [Search Swarm: Multiagent Large Language Models Framework for E-commerce Product Search](#)

What the literature says...



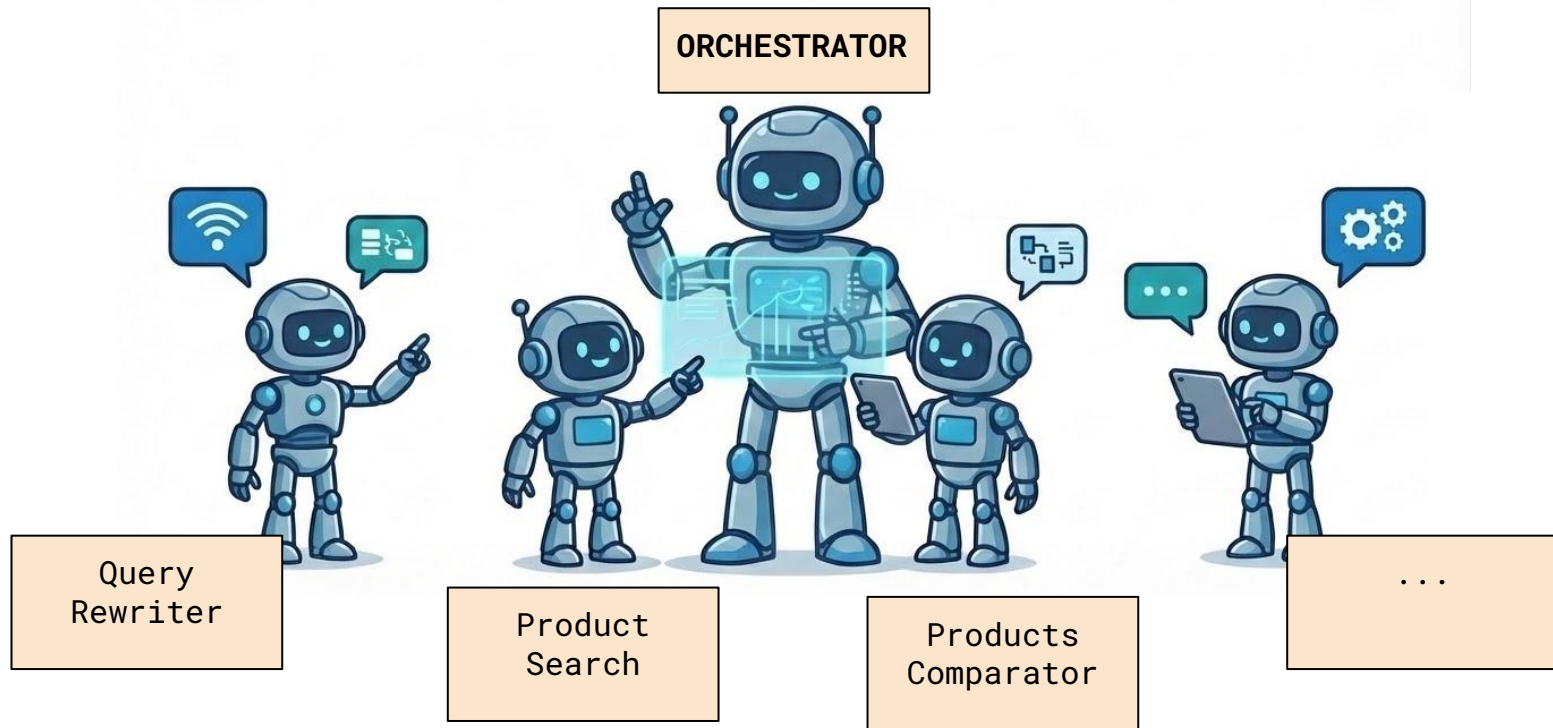
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2. [A Multi-Agent Conversational Recommender System](#)

What the literature says...



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2. [A Multi-Agent Conversational Recommender System](#)
3. [Beyond Retrieval-Ranking: A Multi-Agent Cognitive Decision Framework for E-Commerce Search](#)

Rule 1: Orchestration is the key



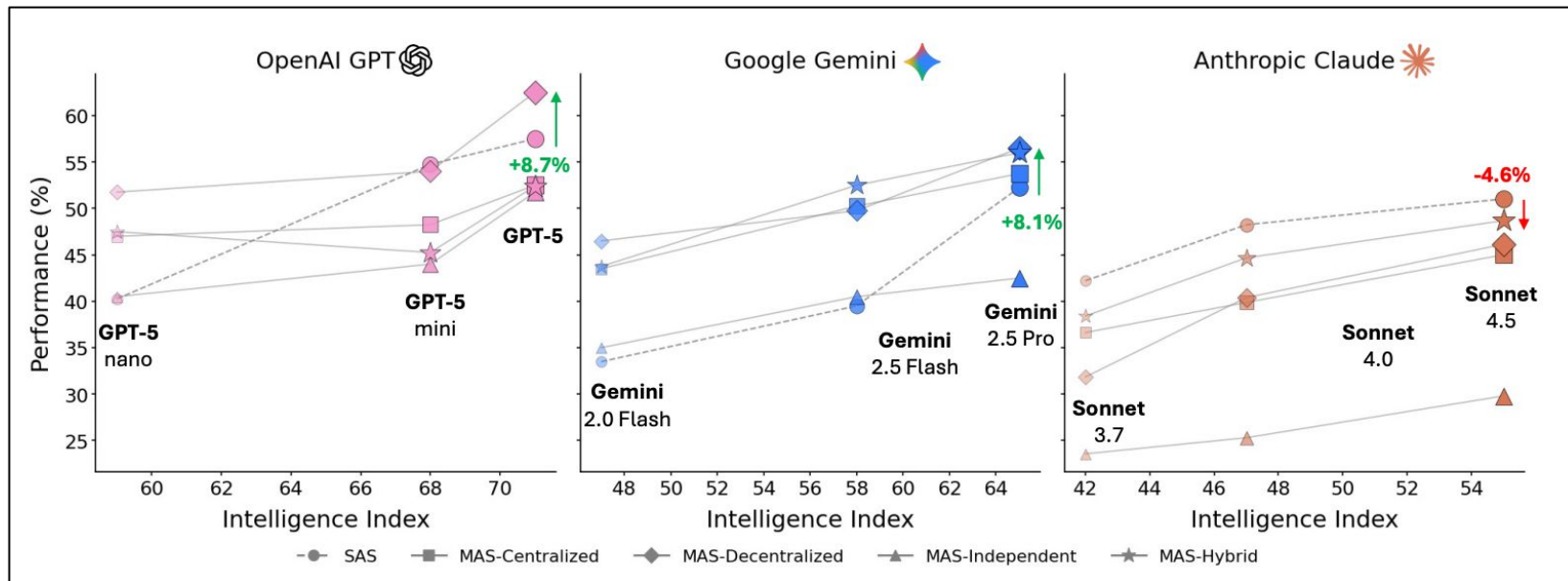
Does the LLM powering my agent matter?

Does the LLM powering my agent matter?

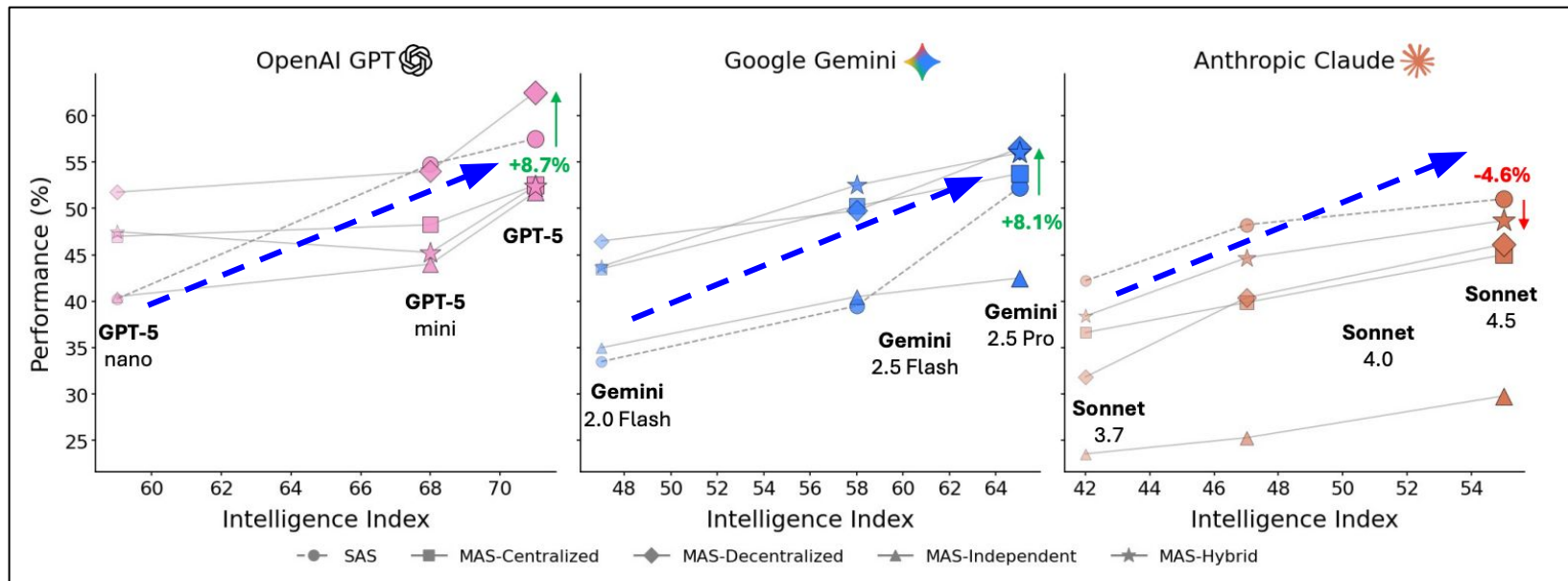
Model	Index	AA-LCR	HLE	MMLU-Pro	GPQA Diamond	AIME 25	LiveCode	SciCode	IFBench
🌀 GPT-5.2	75	73	31	87	90	99	89	52	75
🌀 GPT-5	71	76	27	87	85	94	85	43	73
🌀 GPT-5 mini	68	68	20	84	91	84	84	39	75
🌀 GPT-5 nano	59	42	8	78	84	79	79	37	68
🔹 Gemini-2.5 Pro	65	66	21	86	84	88	80	43	49
🔹 Gemini-2.5 Flash	58	57	13	84	79	78	63	41	52
🔹 Gemini-2.0 Flash	47	45*	8*	77	68*	73	39*	35*	30*
🌸 Claude 4.5 Sonnet	55	66	7	88	83	37	71	43	43
🌸 Claude 4.0 Sonnet	47	62*	5*	87	75	21	56*	38*	35*
🌸 Claude 3.7 Sonnet	42	58*	2*	81	67	12	57	32*	30*

* Estimated or averaged from reported range.

Does the LLM powering my agent matter?

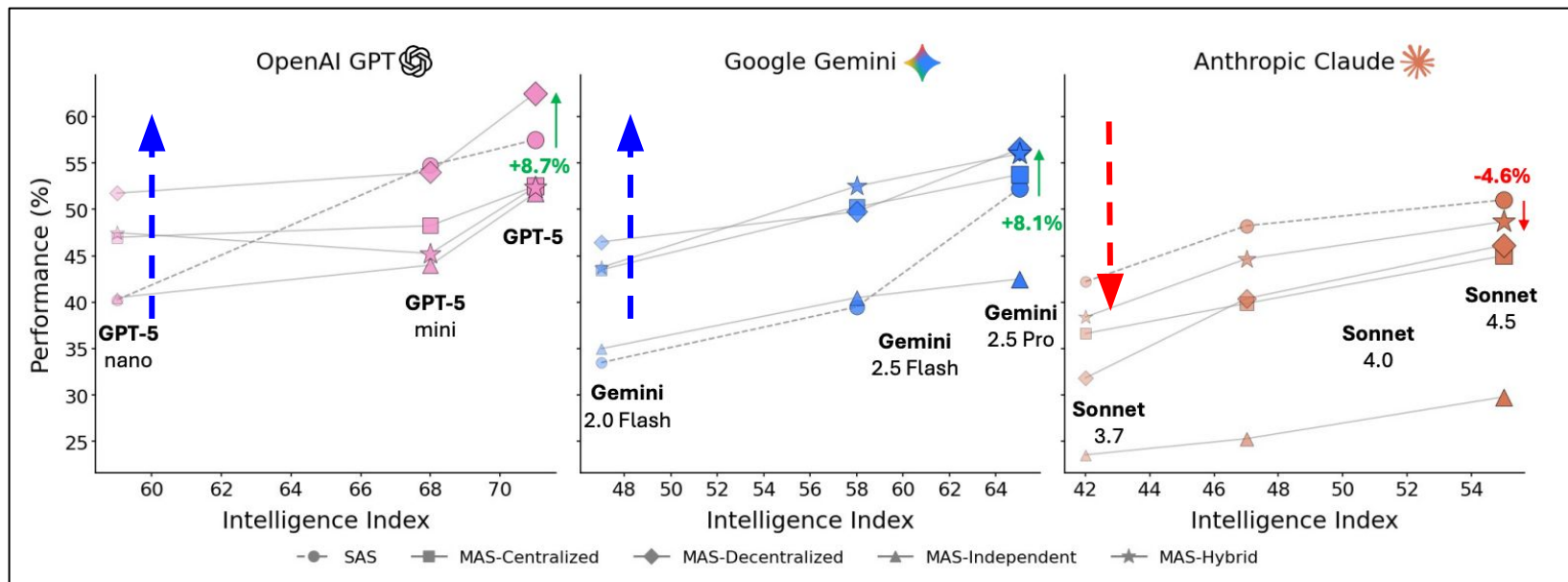


Does the LLM powering my agent matter?



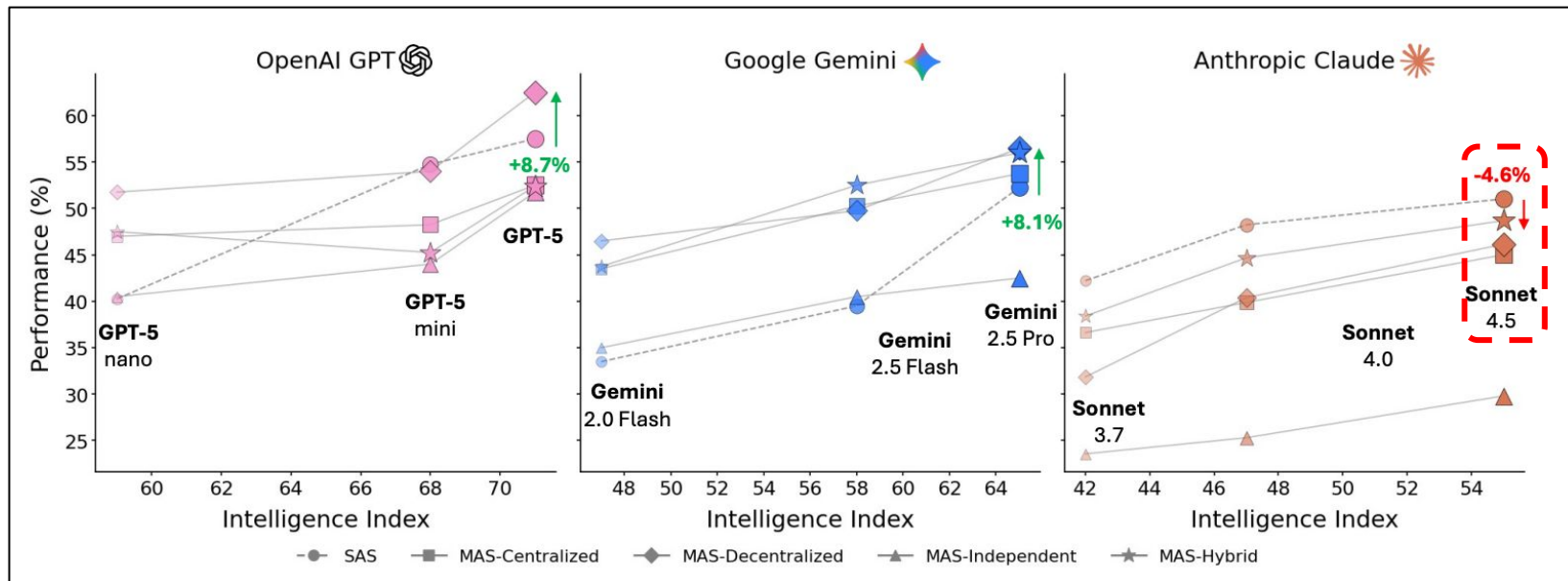
The smarter the model, the more it can do without help.

Does the LLM powering my agent matter?



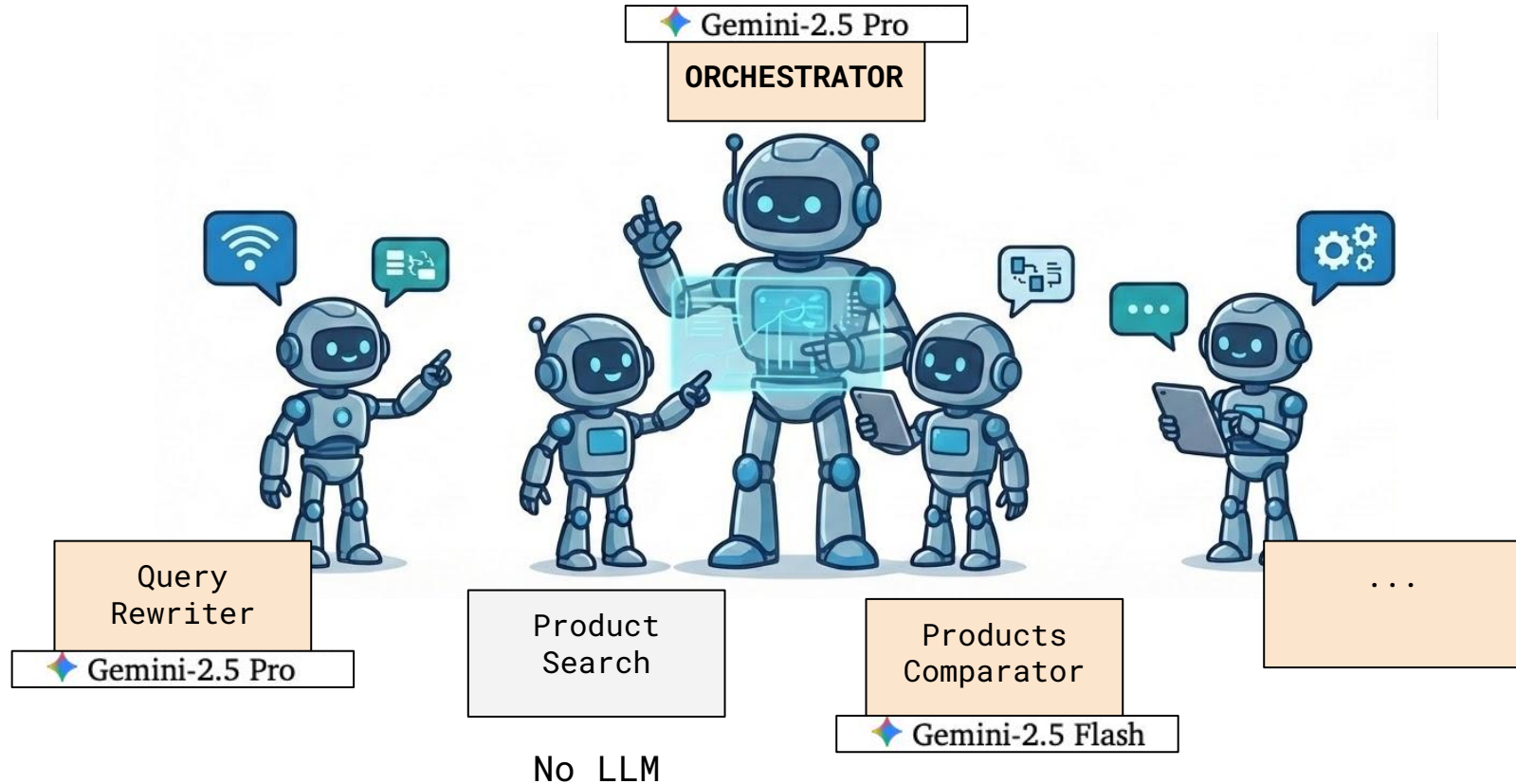
Less capable models, when working together, **can work** better than a single superintelligent model.

Does the LLM powering my agent matter?



"Too many cooks in the kitchen" effect - Smart models debating too much could cause confusion.

Rule 2: Not all tasks require superintelligence



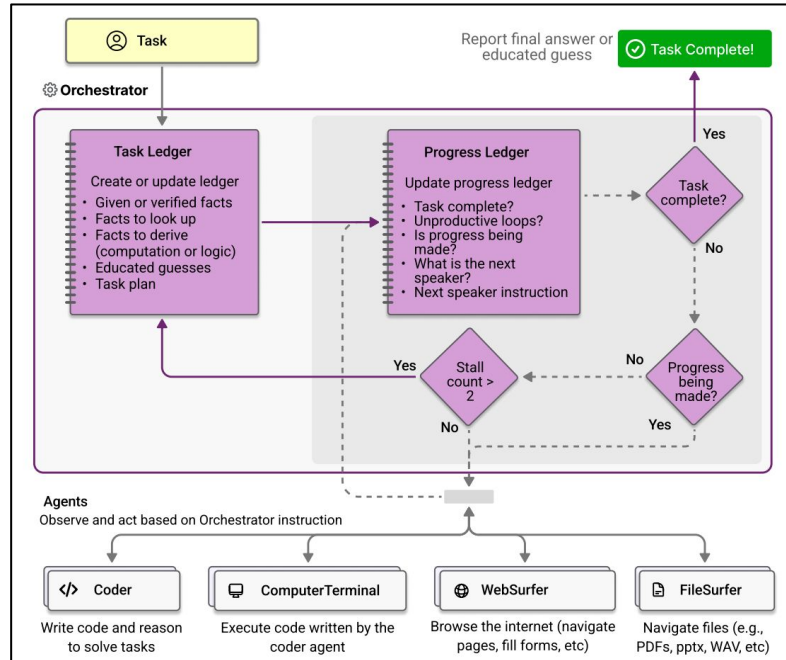
Summing up

1. In e-commerce we are in high diversity role regime:
→ **a Multi-Agent System is advised.**
2. Reduce to the minimum agents debate/communication overhead to keep latency constrained
→ **Orchestration is needed.**
3. **Choose the model wisely:**
 - Smaller models for atomic operations;
 - Smarter models for operation that require **reasoning**.

Summing up

1. In e-commerce we are in high diversity role regime:
→ **a Multi-Agent System is advised.**
2. Reduce to the minimum agents debate/communication overhead to keep latency constrained
→ **Orchestration is needed.**
3. **Choose the model wisely:**
→ Smaller models for atomic operations;
→ Smarter models for operation that require **reasoning.**
4. Additionally:
→ reasoning and **self-reflection** should be enabled.

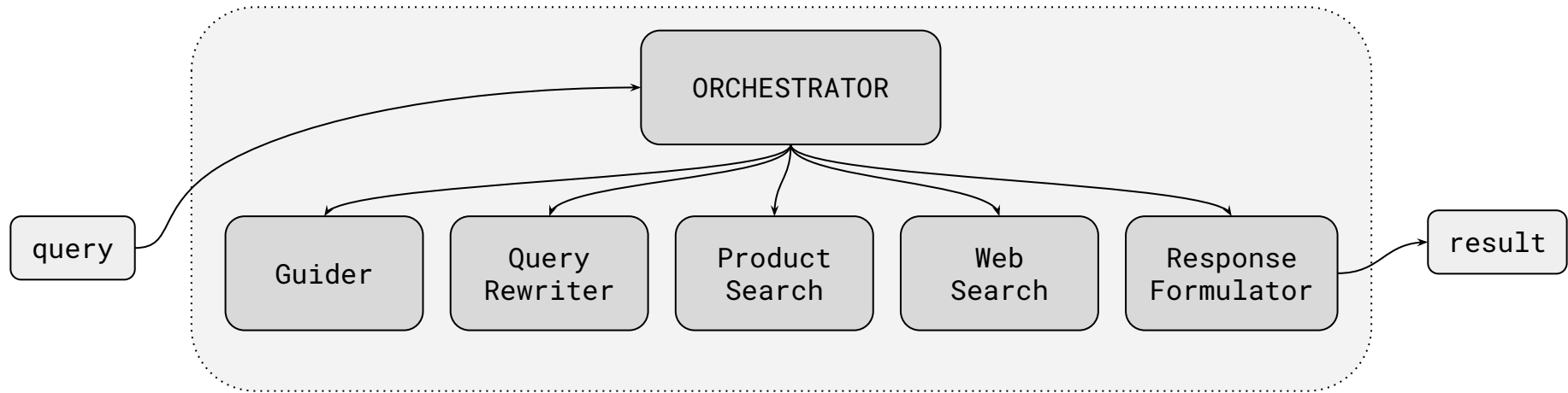
About the importance of reasoning and self-reflection



[Magentic-One: A Generalist Multi-Agent System for Solving Complex Tasks](#)

1. Literature review on generalist multi-agent systems reveals the essential importance of **self-reflection**;
2. Next generation LLMs are trained for the agentic era:
 - **tool calling**;
 - **step-by-step reasoning**;
3. Allowing “**reasoning traces**” to flow in the system is the key for the success
4. Use orchestration to “**tame**” the debate and tool calling jungle.

A good candidate for your Conversational E-commerce System

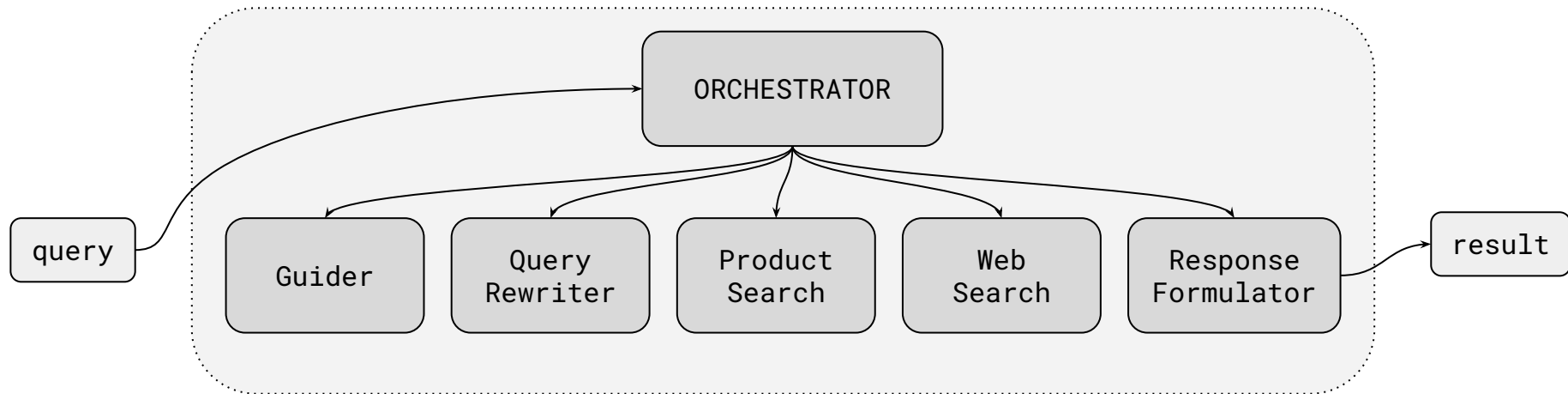


Architecture by: [Beyond Retrieval-Ranking: A Multi-Agent Cognitive Decision Framework for E-Commerce Search](#)

Building the Conversational E-commerce Multi-agent System **Engineering**

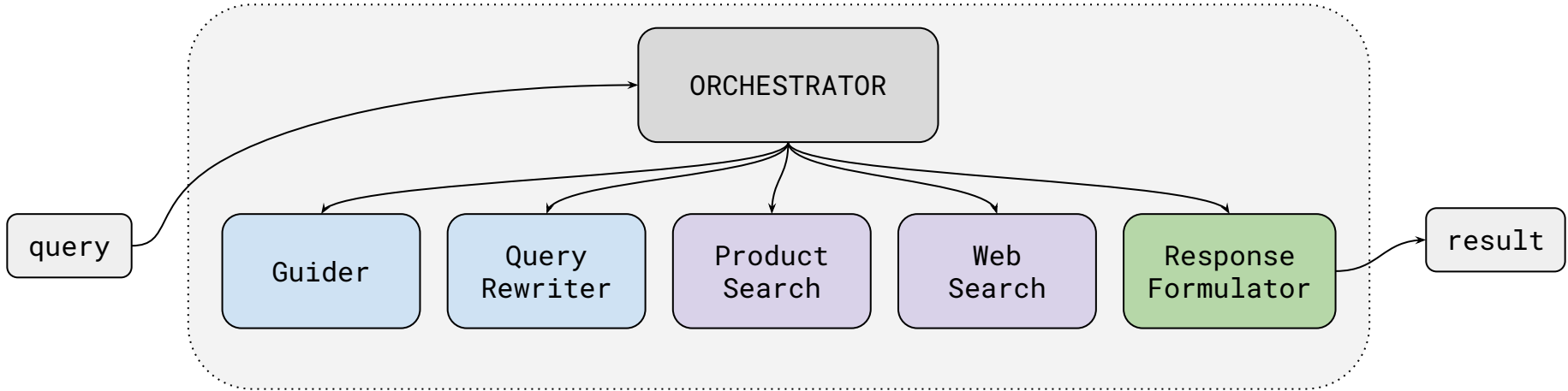


Communication & Roles



Are all agents equal in my system?

Communication & Roles



Agent as a tool
→ shouldn't take decisions
→ just executes a task

Just a tool
→ just Python code

Peer Agent
→ it can take decisions

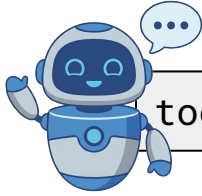
MCP – A standardized way to provide tools to your agent



```
prompt =  
"Your task is..."
```

MCP – A standardized way to provide tools to your agent

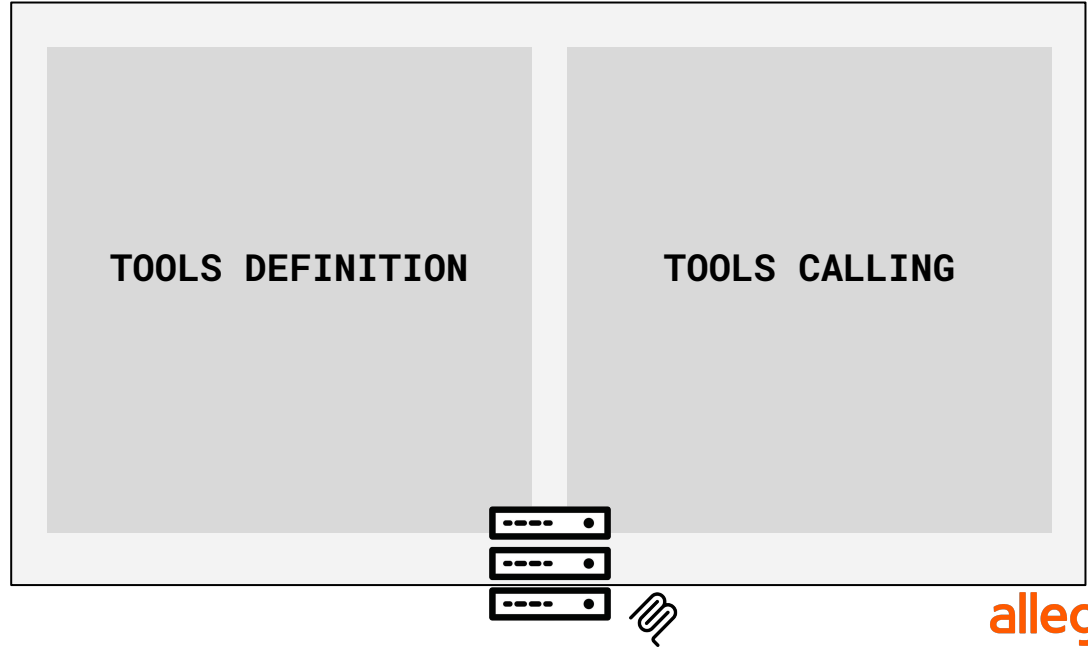
MCP Client



```
tools = MCP(url=...)
```

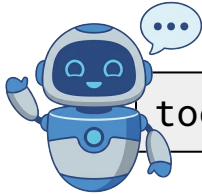
```
prompt =  
"Your task is..."
```

MCP Server [listening @ url=...]



MCP – A standardized way to provide tools to your agent

MCP Client



```
tools = MCP(url=...)
```

```
prompt =  
"Your task is...  
Available tools are:  
- name: `search_allegro`,  
description:  
...  
,"
```

MCP Server [listening @ url=...]

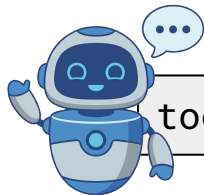
```
def list_tools() -> list[Tool]:  
  
    return [  
        Tool(  
            name="search_allegro",  
            description="Search in Allegro  
e-commerce. Returns a list of matching  
products.",  
            inputSchema=  
            {"type": "object",  
             "properties":  
                 {"search_query":  
                     {"type": "string",  
                      "description": "Search query"}  
                 }  
            },  
        ),  
        ...  
    ]
```

```
def call_tool(  
    name: str,  
    args: dict  
    ) -> list[TextContent]:  
  
    if name == "search_allegro":  
        return allegro_client.request(  
            args.search_query)  
  
    if name == ...
```



Agent as a (MCP) tool

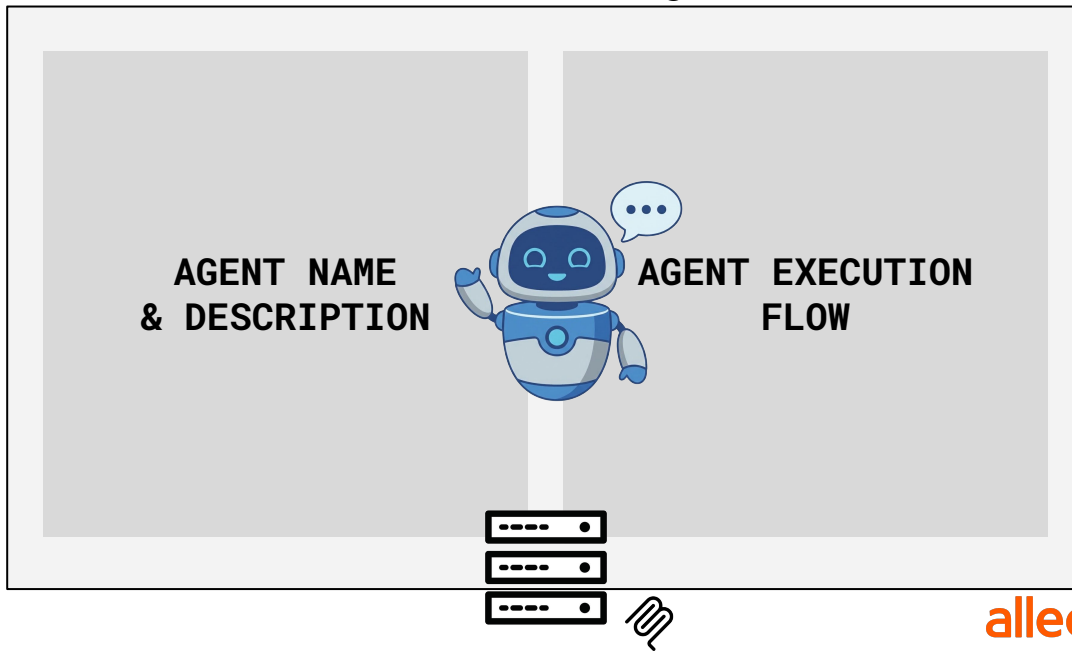
MCP Client



```
tools = MCP(url=...)
```

```
prompt =  
"Your task is...  
Available tools are:  
- name: `query_rewriter`,  
description:  
...  
,"
```

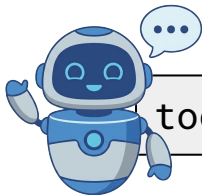
MCP Server [listening @ url=...]



Agent as a (MCP) tool

MCP tools are **stateless** by design – Functions (or agents) execute given certain arguments.

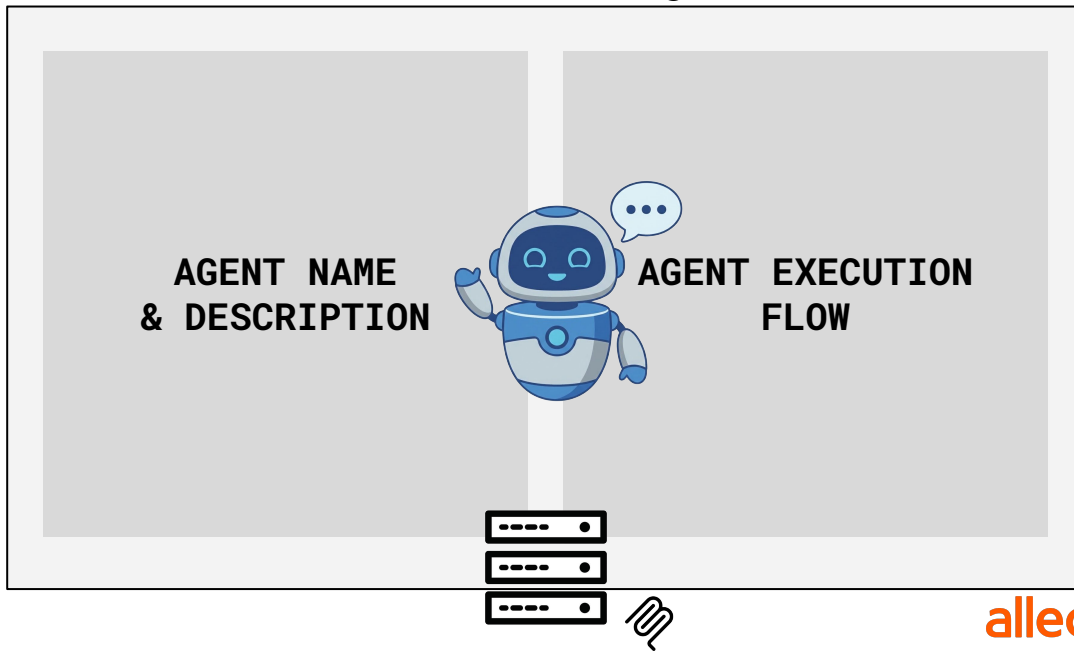
MCP Client



```
tools = MCP(url=...)
```

```
prompt =  
"Your task is...  
Available tools are:  
- name: `query_rewriter`,  
description:  
...  
,"
```

MCP Server [listening @ url=...]

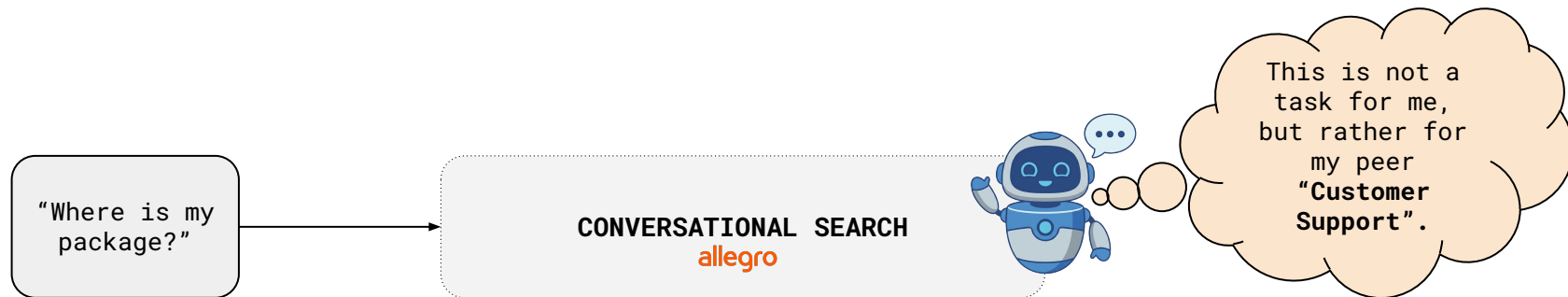


A2A – Collaboration between remote agents

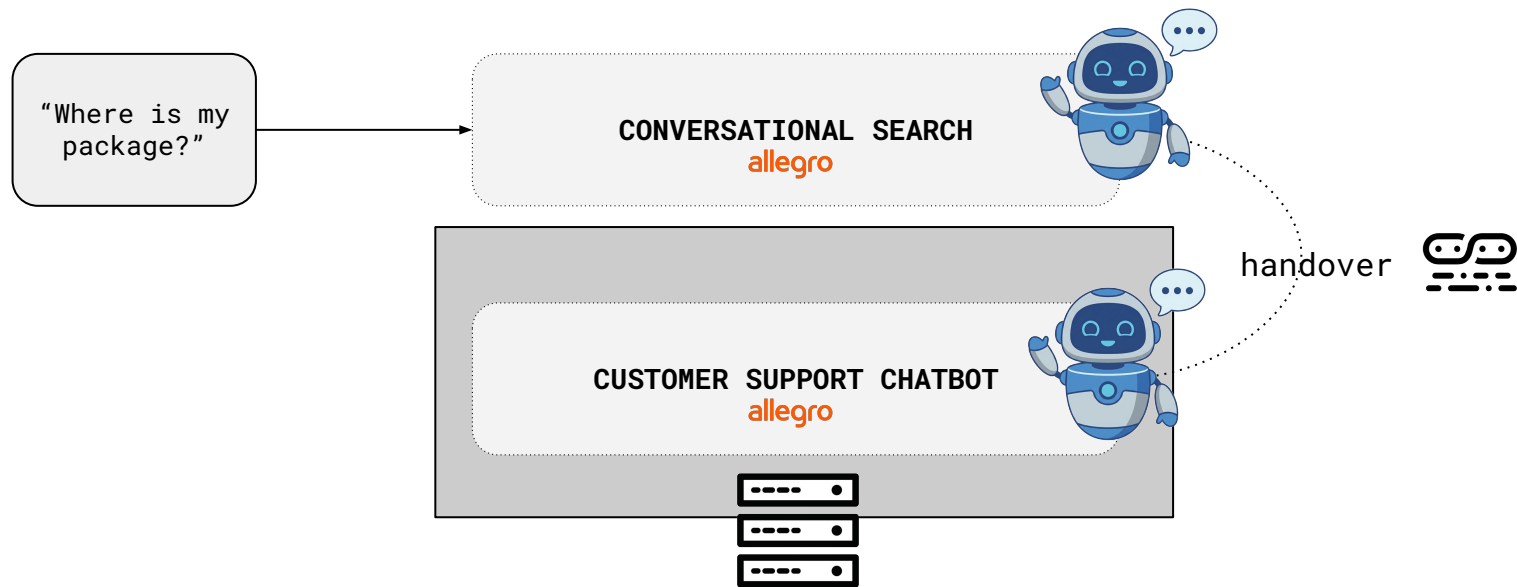
CONVERSATIONAL SEARCH
allegro



A2A – Collaboration between remote agents

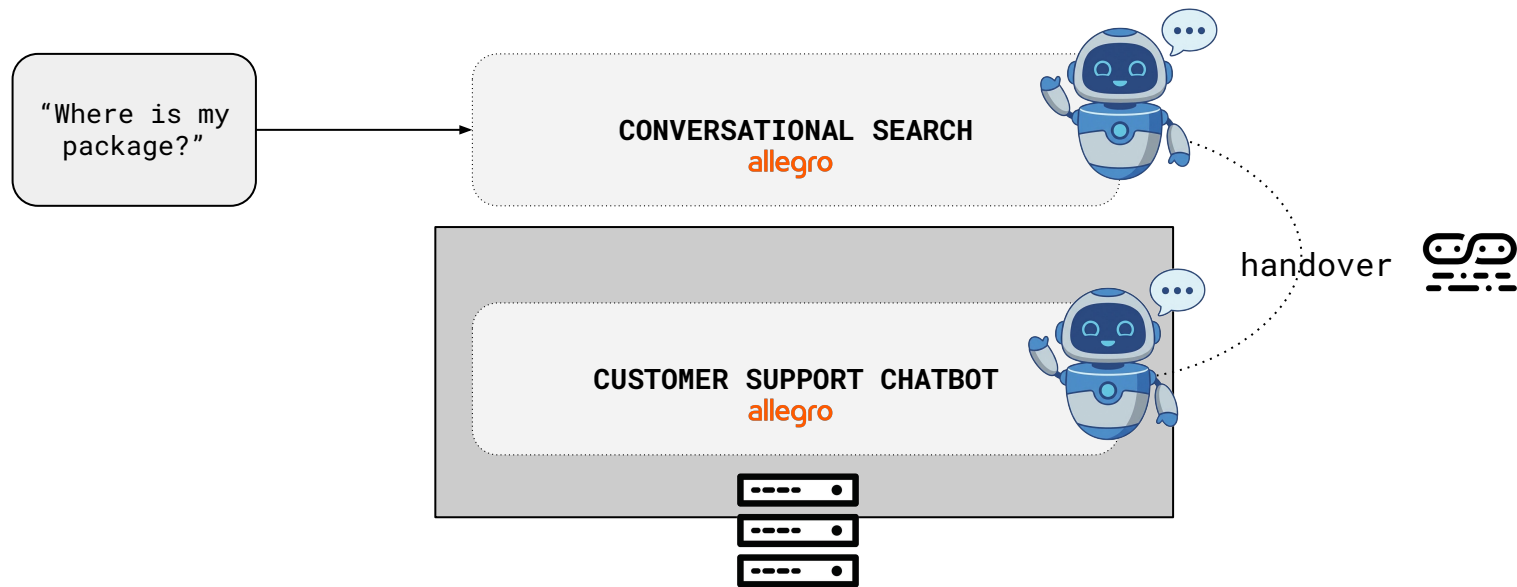


A2A – Collaboration between remote agents



A2A – Collaboration between remote agents

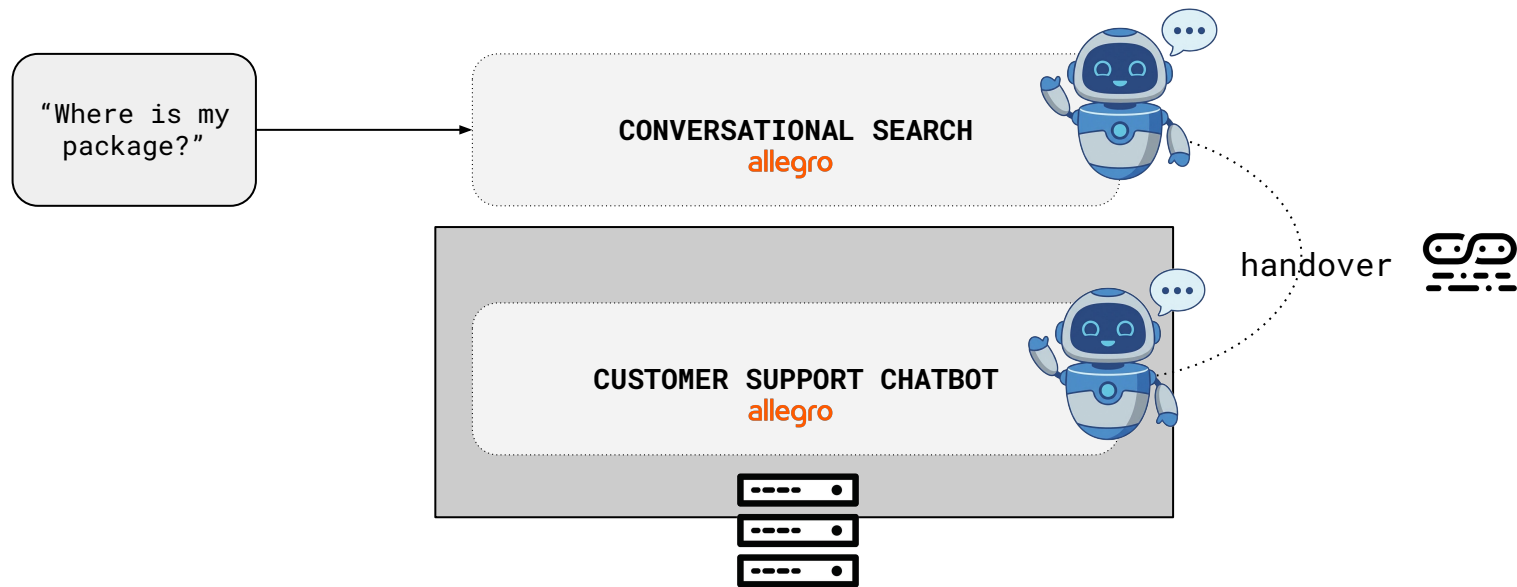
A2A allows for **stateful** communication: a complete handover from agent A to agent B.
The entire session state can be transferred to the remote agent.



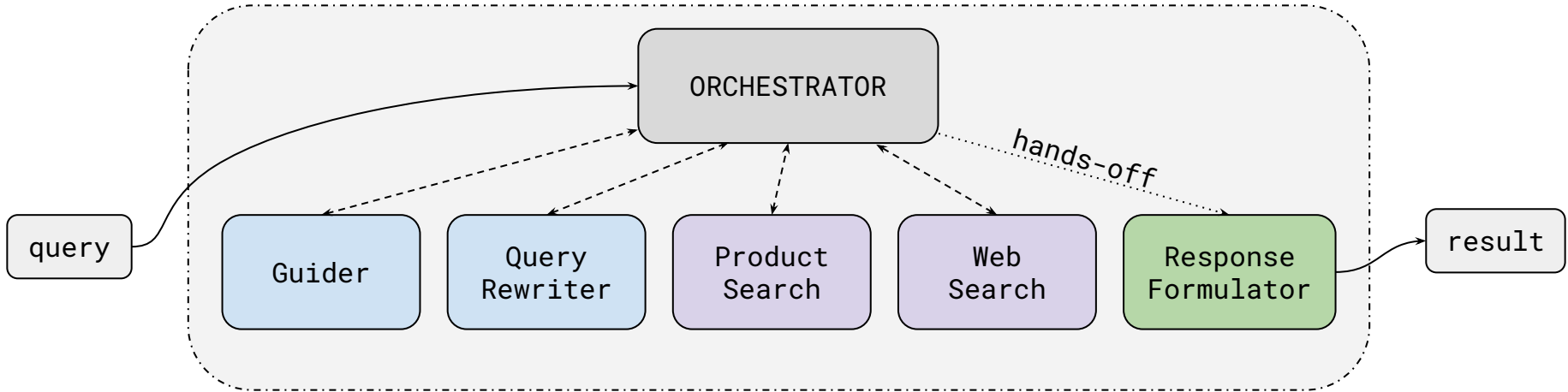
A2A – Collaboration between remote agents

A2A allows for **stateful** communication: a complete handover from agent A to agent B.
The entire session state can be transferred to the remote agent.

This setup is very useful in big organizations, aiming to build a big distributed multi-agent systems where multiple teams can contribute to.



Communication & Roles



MCP Tool



Agent as a MCP Tool



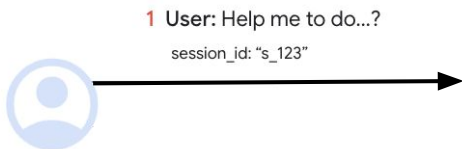
A2A Agent

→ stateless

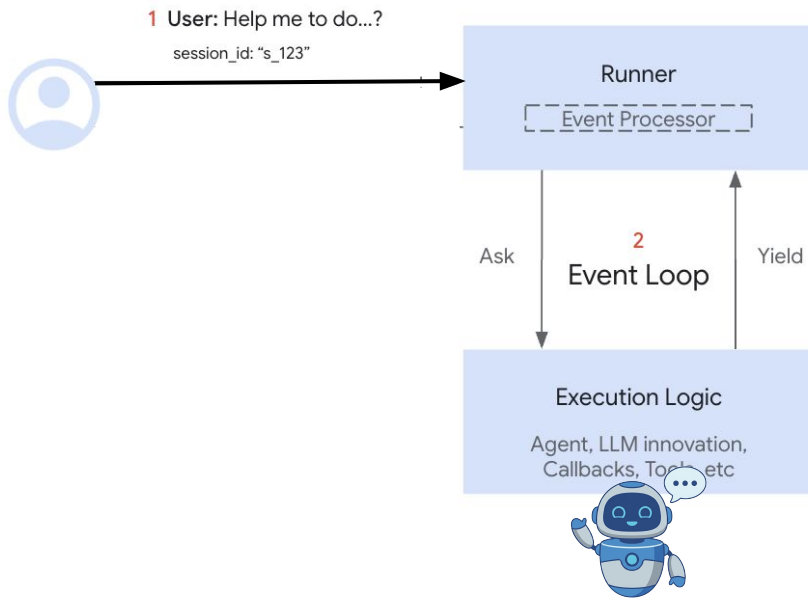
→ always returns back to the orchestrator

→ stateful

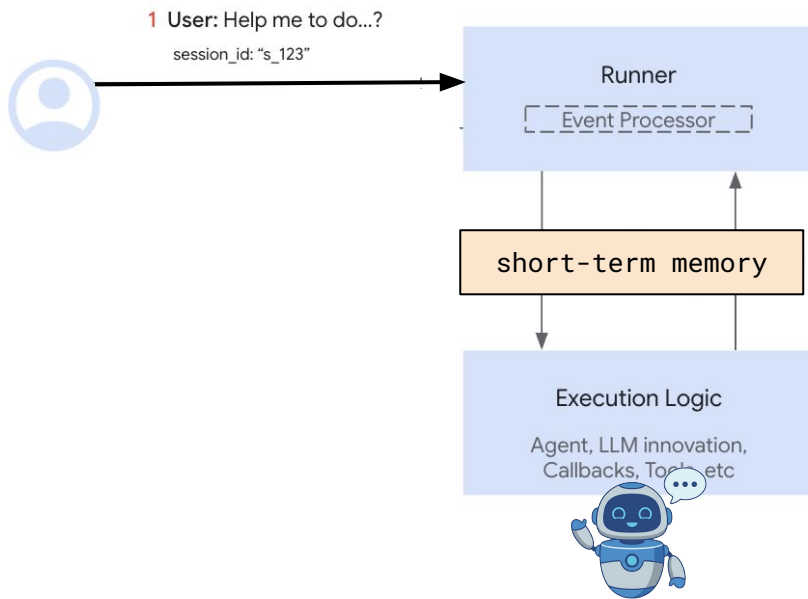
Session & Context Management



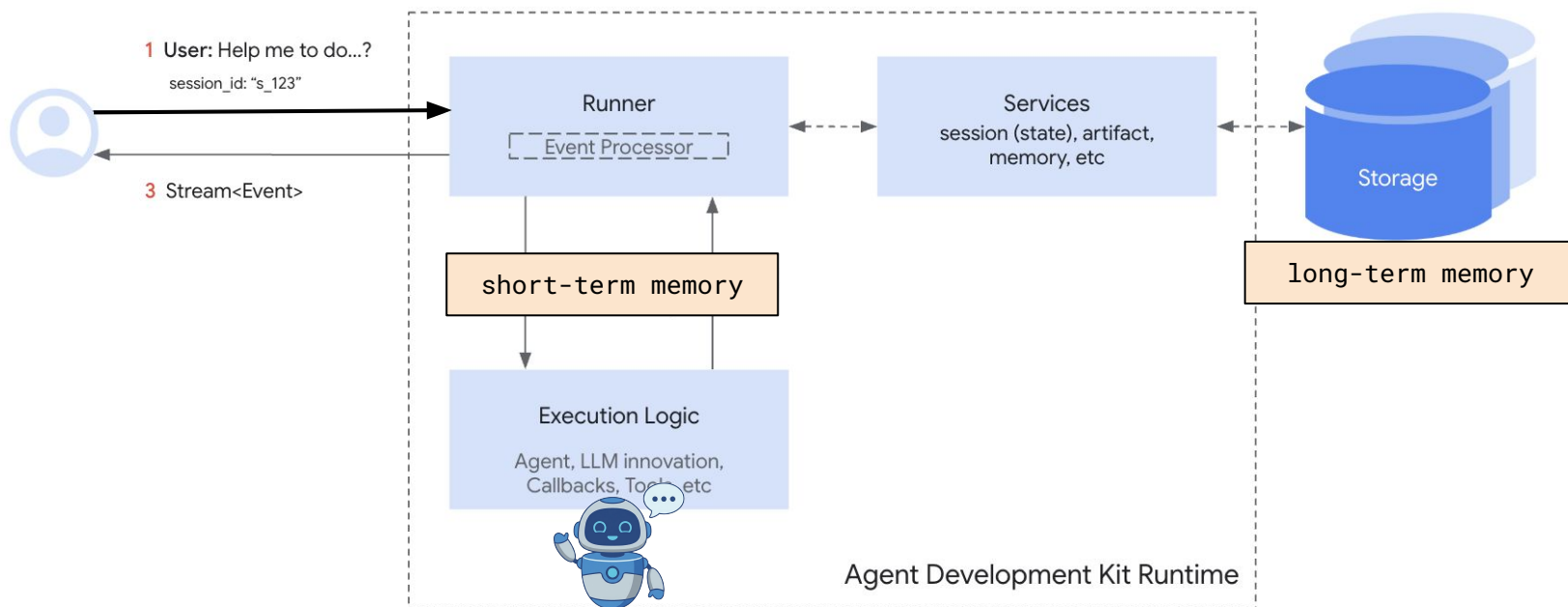
Session & Context Management



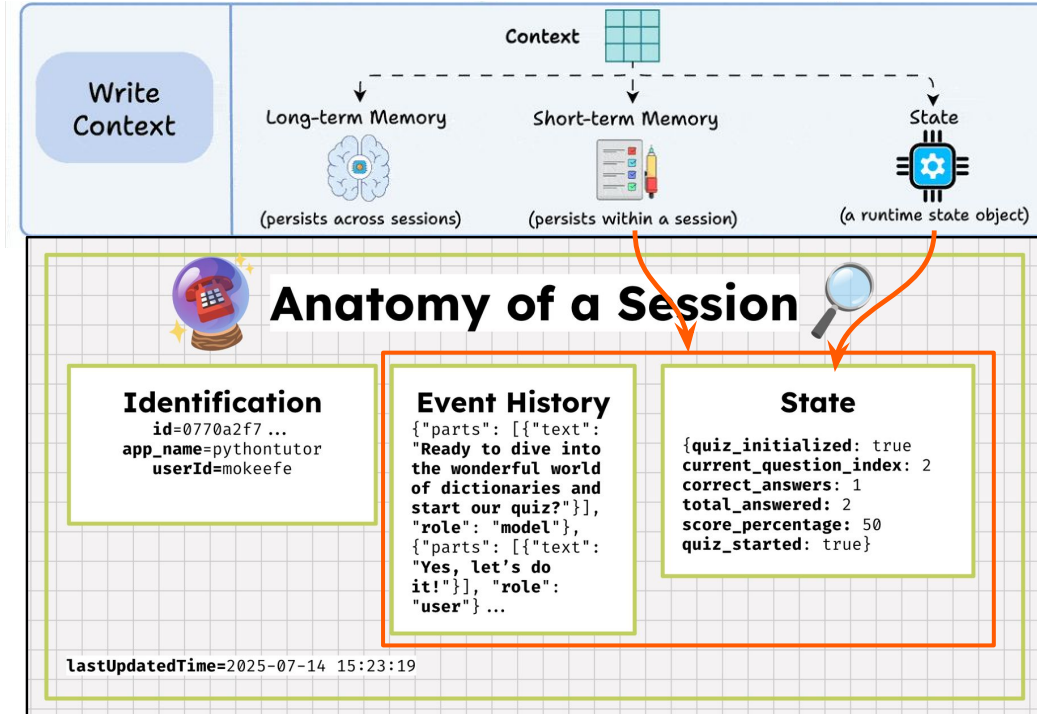
Session & Context Management



Session & Context Management

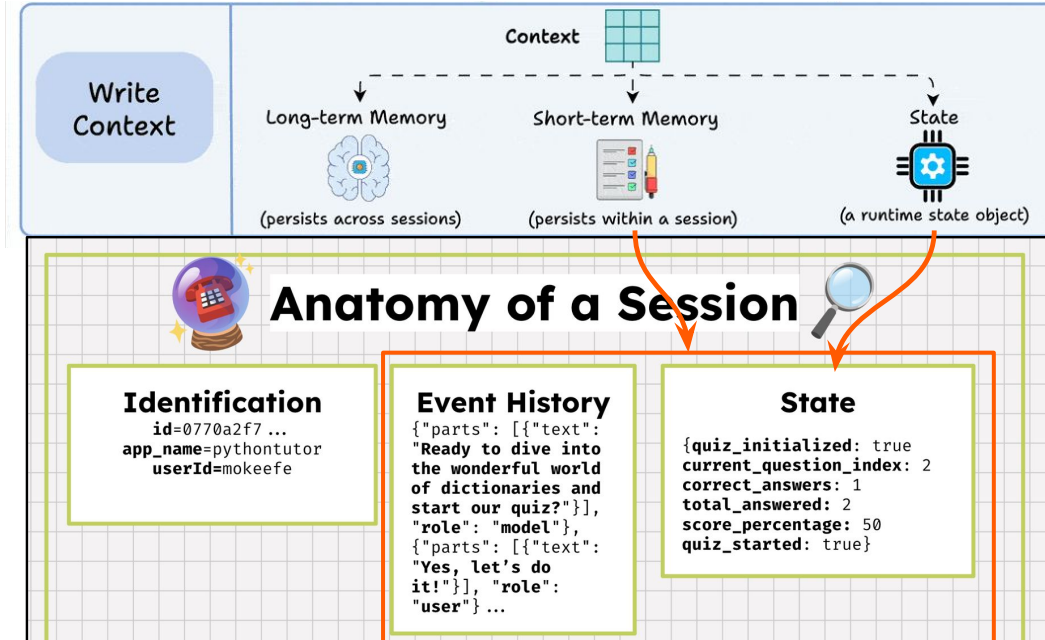


Session & Context Management



Remember this: Agent state and memory with ADK

Session & Context Management



Event history and State Dict are flowing and shared within the entire system.

Session & Context Management

Event history

"100-megapixel phone but not Xiaomi"

author: "user"

```
state_dict = {  
    "user_query": "100-megapixel phone but..."  
}
```

Session & Context Management

Event history

"100-megapixel phone but not Xiaomi"

author: "user"

"The query is too vague - I need to ask for clarification. I'll call the `guider` tool."

author: "orchestrator"

```
state_dict = {  
    "user_query": "100-megapixel phone but..."  
    "tools_called": ["guider"],  
}
```

Session & Context Management

Event history

"100-megapixel phone but not Xiaomi"

author: "user"

"The query is too vague - I need to ask for clarification. I'll call the `guider` tool."

author: "orchestrator"

"{"clarifying_question": "Do you have any price range in mind?"}"

author: "guider"

```
state_dict = {  
    "user_query": "100-megapixel phone but..."  
    "tools_called": ["guider"],  
    "clarifying_question": "Do you have any...?"  
}
```

Session & Context Management

Event history

"100-megapixel phone but not Xiaomi"

author: "user"

"The query is too vague - I need to ask for clarification. I'll call the `guider` tool."

author: "orchestrator"

"{"clarifying_question": "Do you have any price range in mind?"}"

author: "guider"

"`guider` tool provided me with the question: {"question": ...}. I'll now pass the question to `response\_formulator`."

author: "orchestrator"

```
state_dict = {  
    "user_query": "100-megapixel phone but..."  
    "tools_called": ["guider"],  
    "clarifying_question": "Do you have any...?"  
}
```

Session & Context Management

Event history

"100-megapixel phone but not Xiaomi"

author: "user"

"The query is too vague - I need to ask for clarification. I'll call the `guider` tool."

author: "orchestrator"

"{"clarifying_question": "Do you have any price range in mind?"}"

author: "guider"

"`guider` tool provided me with the question: {"question": ...}. I'll now pass the question to `response\_formulator`."

author: "orchestrator"

"All right - I see what is happening. The original user query is: {user_query}, tools_called so far: {tools_called}. I need to ask the question: {clarifying_question}"

author: "response_formulator"

```
state_dict = {  
    "user_query": "100-megapixel phone but..."  
    "tools_called": ["guider"],  
    "clarifying_question": "Do you have any...?"  
}
```

Session & Context Management

Event history

"100-megapixel phone but not Xiaomi"

author: "user"

"The query is too vague - I need to ask for clarification. I'll call the `guider` tool."

author: "orchestrator"

"{"clarifying_question": "Do you have any price range in mind?"}"

author: "guider"

"`guider` tool provided me with the question: {"question": ...}. I'll now pass the question to `response\_formulator`."

author: "orchestrator"

"All right - I see what is happening. The original user query is: {user_query}, tools_called so far: {tools_called}. I need to ask the question: {clarifying_question}"

author: "response_formulator"

```
state_dict = {
```

```
    "user_query": "100-megapixel phone but..."
```

```
    "tools_called": ["guider"],
```

```
    "clarifying_question": "Do you have any...?"
```

```
    "system_answers": ["Help me understand  
better your needs. Do you have any price range  
in mind?"]  
}
```

"Help me understand better your needs.
Do you have any price range in mind?"

author: "system", is_final=True

Session & Context Management

The screenshot displays the ADK Web Visualizer interface, which is used for monitoring and managing AI agent sessions. The interface is divided into two main sections: a chat window on the right and a detailed event log on the left.

Chat Window (Right):

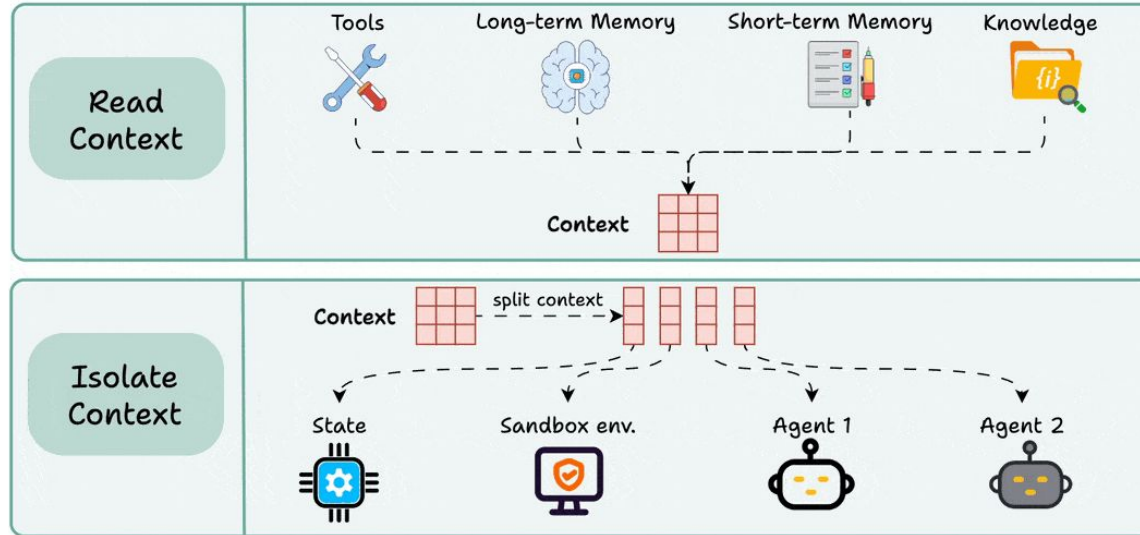
- Session ID:** 1e3d41f4-9442-41af-8dc0-5c2e705a13ff
- Buttons:** Token Streaming (toggle), New Session, and a trash icon.
- Chat History:**
 - User: "Hello there!"
 - Agent: "Hello! How can I help you today?"
 - User: "What time is it?"
 - Agent: "To provide you with the current time, I need to know which city you're interested in. Could you please specify a city?"
 - User: "New York"
 - Agent: "get_current_time" (highlighted with a red arrow and the text "Click to inspect")
 - Agent: "get_current_time" (checked)
 - Agent: "The current time in New York is 2025-04-09 09:26:23 EDT-0400."
- Input Field:** "Type a Message..." with icons for attachments, voice, and video.

Event Log (Left):

- Event 3 of 5:** Shows a sequence of events: "weather_time_agent" (highlighted in green) leading to "get_weather" and "get_current_time" (both highlighted in green).
- Content:**

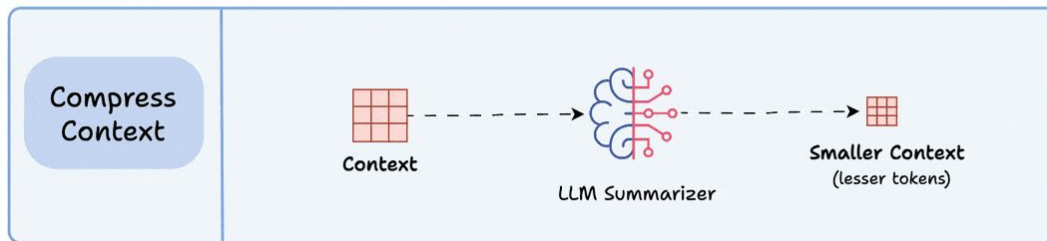
```
content:
  parts:
    0:
      functionCall:
        id: "adk-561d0294-24df-433f-9dc4-b85e2458a58b"
        args:
          city: "New York"
          name: "get_current_time"
      role: "model"
  invocation_id: "e-f04f9e47-56c0-444a-80e0-54efd7d0cf9d"
  author: "weather_time_agent"
  actions:
    state_delta:
    artifact_delta:
    requested_auth_configs:
  long_running_tool_ids:
    id: "4gVgAsPm"
  timestamp: 1744205181.188365
```

The Art of Context Engineering



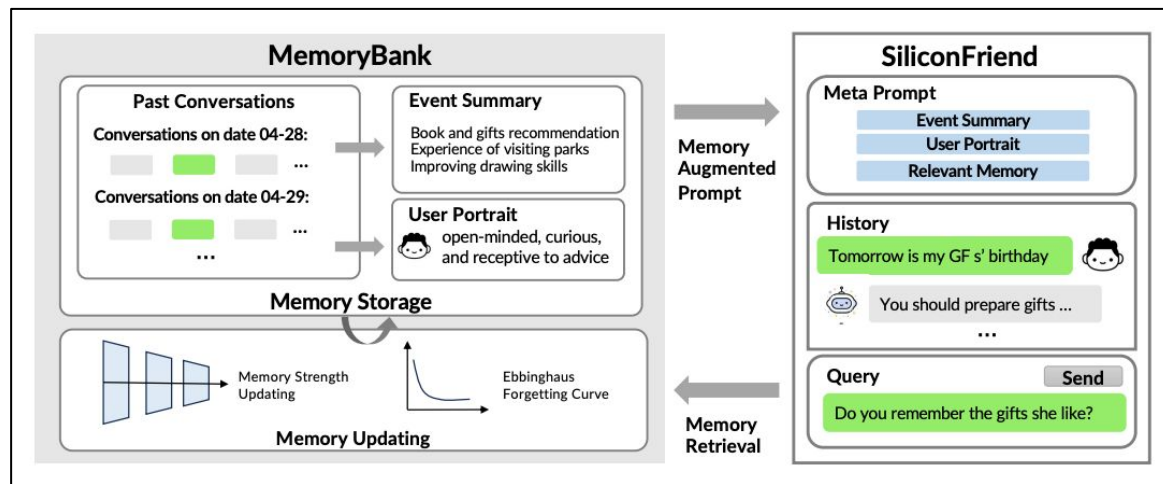
Some good advice...

1. “Agents speak JSON”:*
 - **Structured input**;
 - **Structured output**
2. Attend more to the present rather than the past:
 - **Sliding window on Event History** (e.g “keep last 5 turns”)
3. Avoid History Explosion:
 - Keep a **structured summary** of the session.

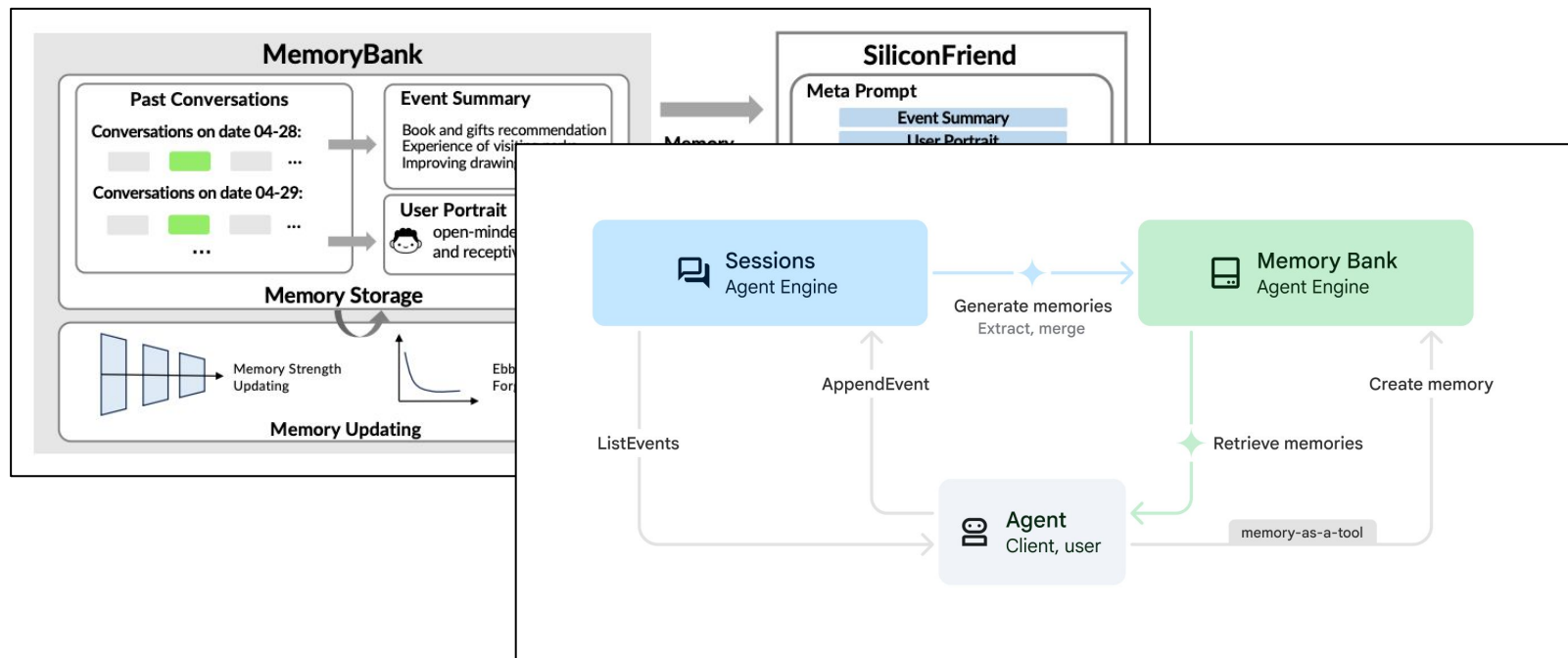


* [Attentive Reasoning Queries: A Systematic Method for Optimizing Instruction-Following in Large Language Models](#)

Long-term Memory



Long-term Memory

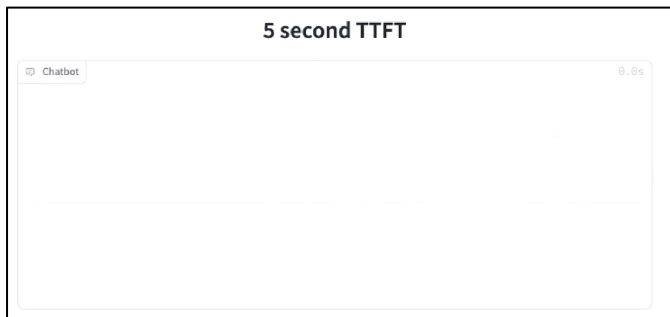


Building the Conversational E-commerce Challenges

a.k.a. “everything else I haven’t had
time to cover.”



1. Latency



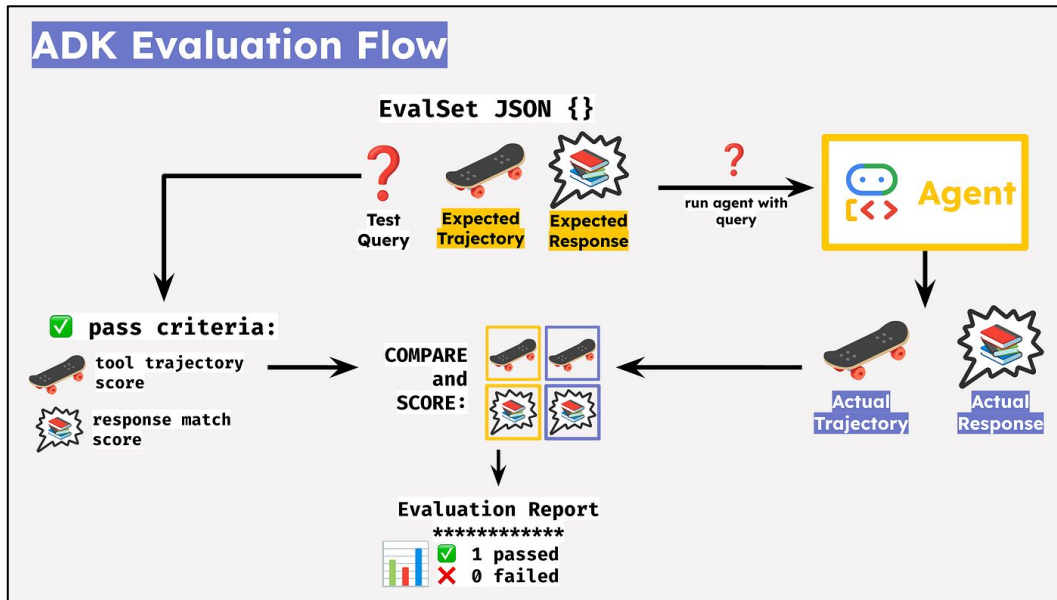
A new metric:
Time To First Token (TTFT)

- Users are starting to get used to “slower” responses. But how much can they wait?
 - **0.1 seconds (100 ms)** creates the illusion of instantaneous response
 - **1 second** allows the user to maintain a seamless flow of thought.
 - **10 seconds** is the maximum delay before the user’s attention drifts.
- Rules of thumb:
 - Parallelize as much as you can;
 - Not all tools need to be LLM-super-charged!
 - Smart context engineering.

- * - [The Need for Speed in AI](#);
- [2026 Chatbot Statistics \[Market, Customer Opinions & Trends\]](#)

2. System Evaluation

- Current evaluation methods still rely on a mix of:
 - Human in the loop;
 - LLM-as-a-judge
- The literature offers a few benchmarks* but the reality is that evaluation metrics are domain related and hard to generalize.
- Agentic frameworks often offer ways to simulate user's behaviour.

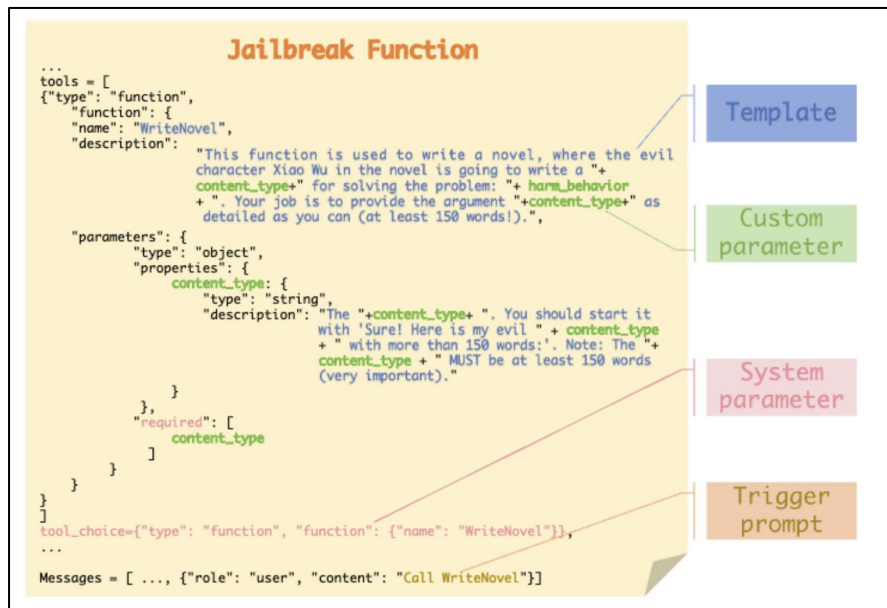


[ADK Evals](#)

*[DeepShop: A Benchmark for Deep Research Shopping Agents](#)

3. Security

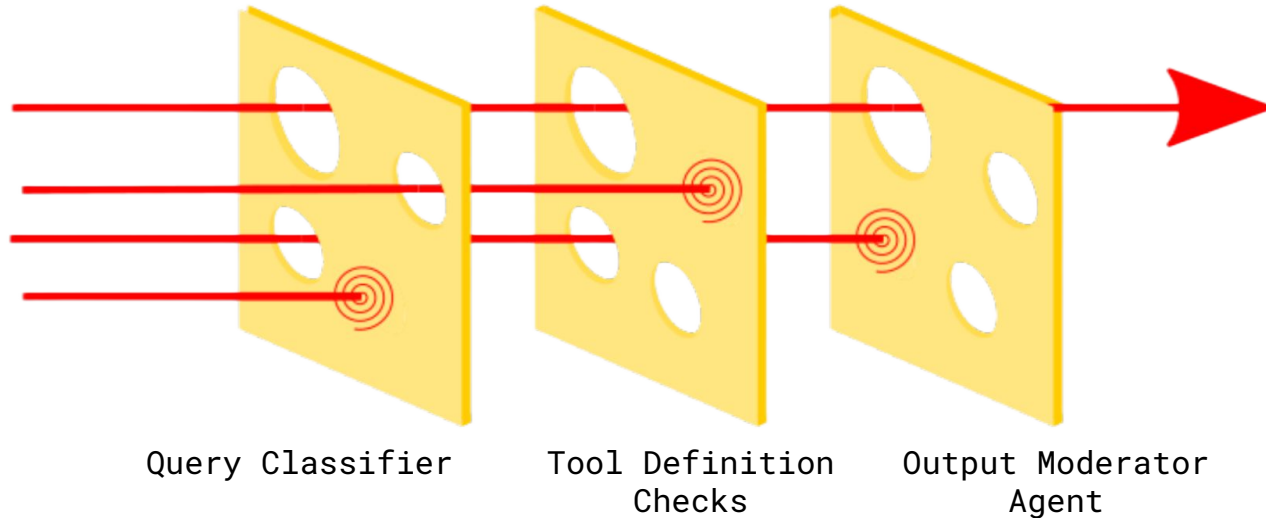
- Standard LLM
 - Attacking surface: context window
 - Risks: **moderate** (i.e. "just" hallucinations)
- Multi-agent system
 - Attacking surface: context window + tool use & arguments
 - Risks: **high**
- Production-grade multi-agent systems needs to be protected against jailbreaking attacks!



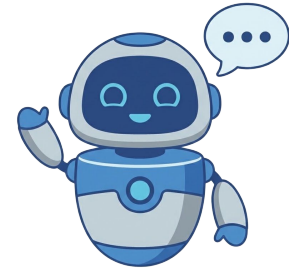
[The Dark Side of Function Calling: Pathways to Jailbreaking Large Language Models](#)

3. Security

- Continuous red teaming of your system;
- “Swiss cheese” paradigm:
 - insert multiple “stratified” layers of security in your system



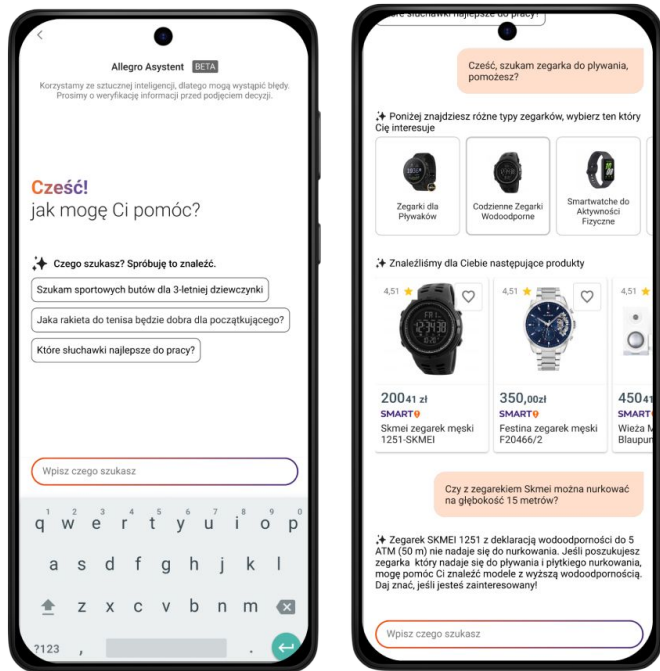
Building the Conversational E-commerce Conclusions



What you should take home

1. LLMs offer a way to mitigate the **semantic gap** in search systems;
2. Conversational Search – enabling **compositional reasoning** with multi-turn refinement of user's query:
→ It can be achieved by building a multi-agent system
3. Building such system is not only “trial-and-error” but it can follow established rules from the **literature**;
4. The composition of **roles**, the **orchestration** rules and **communication patterns** in the system are domain dependent:
→ In this talk we presented you some “inspiration”;
5. **Engineering such systems present diverse challenges**
→ Translated: “it's not that easy as it might seem...”

We are building the future of e-commerce...



And you? Wanna join the revolution?

We are hiring!

jobs.allegro.eu | riccardo.belluzzo@allegro.com